

RESEARCH

Open Access



Machine learning based land use land cover analysis and prediction in Google Earth Engine towards achieving sustainable development in Indian Western Himalayan region

Laxmi Gupta¹ , Jagabandhu Dixit^{2*} , Prem Chandra Pandey^{3*} and Manish Pandey^{4,5}

*Correspondence:
Jagabandhu Dixit
jagabandhu.dixit@snu.edu.in;
jdixitb@gmail.com
Prem Chandra Pandey
prem26bit@gmail.com; prem.
pandey@snu.edu.in

Full list of author information is
available at the end of the article

Abstract

The Indian Western Himalayan (IWH) region experiences persistent threats from natural hazards mainly influenced by land use and land cover (LULC) changes resulting from rapid urbanization. The present study aims to perform machine learning-based LULC classification for the years 2016, 2019, and 2022 and the prediction of future LULC for 2035 using Multi-layer Perceptron Artificial Neural Network-Markov Chain model (MLP-ANN-MCM). LULC map was prepared using Sentinel-2 imagery in Google Earth Engine (GEE), and eight LULC classes were identified with dominance of dense vegetation, scrubland, and barren land. Rapid urban growth and declines in natural habitats were observed from the LULC transition analysis. The kappa statistics validate the accuracy of the model for LULC prediction. Predicted LULC for 2035 indicates an increment in built-up and agricultural areas. The study promotes sustainable development and resilience by emphasizing effective land use planning.

Keywords Sustainable land, Land conversion, Land use land cover, Random forest, MLP-ANN, GEE, Markov chain model

1 Introduction

In recent decades, with population growth and socio-economic development, rapid urban expansion has occurred worldwide [52, 70, 72, 117]. The urban expansion will cause the transformation of agricultural and forested areas into built-up areas, and this significant alteration will pose considerable challenges such as biodiversity loss, urban heat island effects, environmental degradation, increased frequency and intensity of natural hazards, emission of greenhouse gases, and impacting the quality of human life [10, 14, 22, 104, 118, 121]. Land use and land cover (LULC) are entirely different terminologies but very closely interrelated. The term land use primarily refers to the socio-economic attributes of land and has significant implications for ecosystem sustainability, resource availability, quality of life, and the overall functioning of societies [113]. Land



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

cover denotes the natural setting of the land and mainly refers to natural elements such as vegetation, water bodies, and barren land, as well as human-made features like buildings and roads [32]. Both land use and land cover are often studied to understand the spatial distribution and dynamics of human–environment interactions [86].

There are wide LULC applications in geospatial tasks like environmental management, urban planning and development, public health, agriculture, natural resource management, disaster risk management, climate change assessment, and fostering sustainable development [3, 7, 39, 41, 43, 122]. According to various researchers, for sustainable development, it is essential to study the dynamic nature and trend analysis of LULC classes [79, 91, 98, 101, 119]. With the advent of geospatial technologies, satellite imagery and aerial photography provide high-resolution data, and different satellite sensors such as optical, radar, and LiDAR sensors are available to capture data over large geographic areas in a single pass [44, 47, 62]. These geospatial data can be collected repeatedly over time, facilitating the monitoring of temporal changes in LULC dynamics, such as urban expansion, deforestation, and agricultural changes. Remote sensing techniques, including spectral analysis and image classification algorithms, enable automated extraction and classification of land cover information from satellite imagery [80, 107].

For LULC classification studies conducted by Lu et al. [64], Jia et al. [46], Kumar and Acharya [60], Mohajane et al. [67], Alam et al. [8], and Naikoo et al. [71] used Landsat images with continual global coverage at moderate to high resolution. The imageries provided by Sentinel Missions are widely applied in land cover mapping, vegetation studies, hazard mapping, and biodiversity studies due to their better resolutions, increased revisit times, and broad swath width [24, 57, 59, 68, 103, 111]. Different LULC classification and mapping studies involve integrating Landsat and Sentinel data due to improved classification accuracy and enhanced temporal and spatial coverage [26, 42, 73, 77, 94, 108]. The processing and analyzing of satellite data requires high processing power and huge storage capacity, and Google Earth Engine (GEE), a cloud-based platform, satisfies all the requirements for an accurate LULC classification and mapping [38].

Various studies, such as Huang et al. [45], Shetty [96], Zhang et al. [120], and Zhao et al. [123] have utilized GEE for LULC classification and mapping using Sentinel data in different regions globally. GEE provides free satellite imagery from Landsat, Sentinel, MODIS, and many more, and it employs MapReduce architecture for parallel processing, which converts large amounts of data into smaller data, making processing easier across multiple devices [77]. GEE contains different programming libraries such as JavaScript and Python client library, which makes complex modeling algorithms relatively easy, and these algorithms can be applied using the satellite data in various fields such as deforestation monitoring, crop monitoring, water resource management, urban growth and infrastructure monitoring, disaster management and response, ecosystem monitoring and conservation, and climate change studies [25, 38, 114].

The accuracy of LULC classification depends on several factors, such as the source of data, training of datasets, number of classes selected, and classification method [42, 63]. LULC classification methods include machine learning algorithms such as supervised, unsupervised, object-based, and deep-learning classification [50]. The common supervised classification algorithms include Maximum Likelihood, Support Vector Machines (SVM), Random Forest (RF), Classification and Regression Tree (CART), and Artificial Neural Networks (ANN) [9, 10, 23, 123]. The algorithms mentioned above are found

to have high accuracy depending on the objective of the studies. Studies carried out by Shetty [96], Loukika et al. [63], Aldiansyah and Saputra [9], Cengiz et al. [23], and Zhao et al. [123] observed that the Random Forest (RF) classifier is the most suitable one for LULC mapping and classification in the GEE platform using satellite data.

LULC classification and mapping provide an understanding of the present landscape scenario and the area distribution of LULC classes. The LULC of a region is dynamic and changes over time, so modeling and predicting LULC is essential to understand the evolution of landscape over time [54, 73, 85]. It will contribute to advancing scientific understanding of land use dynamics and their drivers by simulating the complex interactions between human activities, environmental factors, and land-use changes [119]. The LULC change analysis and its prediction are applied in studies related to urbanization, climate change, urban planning, development, etc. [35, 53]. Many models are available for change detection and prediction of LULC some of the models are SLEUTH, cellular automaton (CA)–Markov chain model (MCM), artificial neural network/multiple perceptron neural network (ANN/MLPNN)-based model, IDRISI's CAMARKOV, CLUE-S/Dyna-CLUE, DYNAMICS EGO, and Land Change Modeller [51, 110]. Various studies have shown that ANNs can readily discern complex patterns in datasets and effectively represent nonlinear relationships.

Among the many neural networks, the cellular automaton (CA)–Markov chain model (MCM) is most utilized for different applications. Singh et al. [99] applied the Cellular Automata (CA)-Markov Chain Model (MCM) to identify LULC changes in the Allahabad district, India, from 1990 to 2010 and to predict LULC for 2020. Ahmad et al. [5] studied the dynamics of LULC in Ranchi, India, from 1989 to 2015 and predicted the LULC transition matrix for the year 2028 using the cellular automaton (CA)–Markov chain model (MCM). Beroho et al. [18] predicted the LULC of the Mediterranean Watershed in Morocco for 2028, 2038, and 2050 using a CA-Markov forecasting model and found the results satisfactory as estimated by kappa indices. Several studies, Sardar and Samadder [92], Vinayak et al. [113], Redowan et al. [89], and Al-Dousari et al. [10], considered hybrid models, such as the Multi-Layer Perception Neural Network (MLP-NN) model integrated with the MC model for the LULC change detection and prediction studies [36]. MLP-NN consists of three layers, i.e., input, hidden, and output layers, and MLP-NN has the highest power to model transition potential effectively for simulating changes in LULC. This hybrid model produces more accurate results and combines MLP-NN and MC models.

For IWH region, LULC studies have been conducted distinctly for each state, such as Alam et al. [8] performed LULC analysis of the Kashmir region using Landsat satellite data, Taloor et al. [102] studied GIS-based LULC dynamics of Western Doon Valley, Uttarakhand, India, Prashanth et al. [84] carried out analysis of LULC change in the Himachal Himalayan watershed and its implication on soil erosion, Bhasin et al. [19] conducted LULC change detection analysis for trans Himalayan region of Ladakh using remote sensing. These studies were limited to specific regions using conventional remote sensing and GIS techniques. The studies mainly focus on assessing the impact of LULC change on climate change, water resources, soil erosion, urban heat islands, etc. Limited studies were available on the prediction of LULC, especially for the entire IWH region consisting of Ladakh, Jammu and Kashmir, Uttarakhand and Himachal Pradesh, thereby offering limited insights into future land use scenarios crucial for sustainable

development planning. In the present study, four states of the Indian Western Himalayan region, namely Uttarakhand, Himachal Pradesh, Jammu and Kashmir, and Ladakh, are selected as the study area. These regions are considered fragile due to the Himalayan young fold mountains. With the increase in urbanization, these regions are facing rapid developmental activities that are leading to an increase in the frequency of disasters like floods, landslides, flash floods, and biodiversity loss. Assessment of temporal and spatial changes in LULC of these regions becomes crucial to mitigate or minimize the loss caused by the rapid changes in LULC. In the present study, Sentinel-2 data were utilised, and a machine learning-based Random Forest (RF) classification technique was applied in Google Earth Engine (GEE) to prepare the LULC for 2016, 2019, and 2022. Then, temporal and spatial changes of LULC for the years 2016–2019, 2016–2022, and 2019–2022 were analyzed using the modules for land use change evaluation (MOLUSCE) plugin in QGIS. The LULC prediction map for 2035 for the study area was also prepared using the multi-layer perceptron Artificial Neural Network-Markov Chain Model (MLP-ANN-MCM). The results of the present study can assist urban planners and policymakers in developing sustainable strategies to address unplanned urban growth. Understanding the present patterns of LULC and potential future expansion opportunities enables the formulation and execution of various short and long-term development plans for the study area.

1.1 Land use land cover change analysis and implications for sustainable development

Sustainable development relies on alterations in land use and land cover (LULC), which stand out as critical contributors to global environmental shifts [14]. Understanding the status of LULC and its characterization is considered a step toward achieving sustainable development. Land use and land cover are related to the socio-economic and natural attributes of land, therefore, they can be linked to different environmental changes [91]. Any shifting in LULC will impact the natural processes that are taking place on the earth's surface. Eniolorunda et al. [34], Yang et al. [119], Surya et al. [101], Koroso et al. [58], and Arifeen et al. [14] highlight the importance of assessing changes in the landscape pattern that occurs during urbanization in urban planning and sustainability. With economic growth and the increasing needs of the urban population, the quality of the environment is declining due to rising pollution, biodiversity loss, and the depletion of natural resources. Rapid land use conversion occurs due to the pressure rising from population migration and urbanization [16]. At present, the development of any region does not depend solely on economic growth; it is based on sustainable development. Sustainable development involves managing the natural environment to support economic sustainability and enhance living standards, and it also depends on renewable resources to maintain environmental equilibrium [98].

There are 17 Sustainable development Goals (SDGs), including 169 targets proposed by the United Nations, directly or indirectly related to LULC [105]. Out of the 17 goals, the United Nations (UN) Sustainable Development Goals 2, 11, and 13 are directly aligned with the land use land cover changes. Land Use and Land Cover (LULC) are directly related to achieving UN Sustainable Development Goal 2 (SDG 2), which aims to eradicate hunger, achieve food security, improve nutrition, and promote sustainable agriculture; this goal is closely linked to LULC changes [97]. With the population growth and shift towards urban centers, the conversion of agricultural land is taking place,

which can be responsible for reducing arable land for food production [119]. As a result, food production will decline, contributing to food insecurity. Changes in LULC also influence the hydrological cycle, which can impact water quality and availability, thus reducing agricultural productivity [57, 77]. UN Sustainable Development Goal 11 (SDG 11) primarily focuses on making cities and human settlements inclusive, safe, resilient, and sustainable. Changes in LULC often involve expanding urban areas, leading to land consumption, encroachment into green spaces, loss of biodiversity, and fragmentation of ecosystems. With the change in LULC, the vulnerability to natural hazards like floods and landslides also increases [39]. So, mitigation measures must be incorporated into land use planning in hazard-prone areas. LULC changes often accompany infrastructure development projects such as transportation networks, utilities, and public amenities. With the help of sustainable infrastructure planning and design, the environmental impacts will be minimized, and urban resilience to climate change will be enhanced.

2 Study area

The study area extends from 28.73°N to 37.07°N and 72.67°E to 81.02°E as shown in Fig. 1, covering a total area of about 329,511.16 km² contributed by Himachal Pradesh (55,673.1 km²), Uttarakhand (54,035.9 km²), Jammu and Kashmir (53,862.5 km²), and Ladakh (165,939 km²) [40]. The study area is a part of the Hindu Kush-Himalaya (HKH) region, a vast mountainous area in South Asia, covering approximately 4.2 million square kilometres across eight countries: Afghanistan, Bangladesh, Bhutan, China, India, Myanmar, Nepal, and Pakistan [33]. The HKH region is also known as the “Third Pole” of the Earth as the snow and glaciers of the Tien Shan mountains, and the HKH together are the most significant snow and glaciers covered area outside of the North and South Poles [40, 78]. The climate of Uttarakhand and Himachal Pradesh ranges from subtropical in the southern plains to alpine in the northern mountainous regions [4, 40, 61]. The region experiences monsoon rains from June to September, which are vital for agricultural activities but can also lead to landslides and flash floods in hilly areas. Jammu and Kashmir experiences a subtropical climate in Jammu and an alpine climate in the Kashmir Valley and Ladakh [28]. The region experiences a continental climate with distinct seasons, and the Kashmir Valley is known for its picturesque spring bloom. Ladakh is

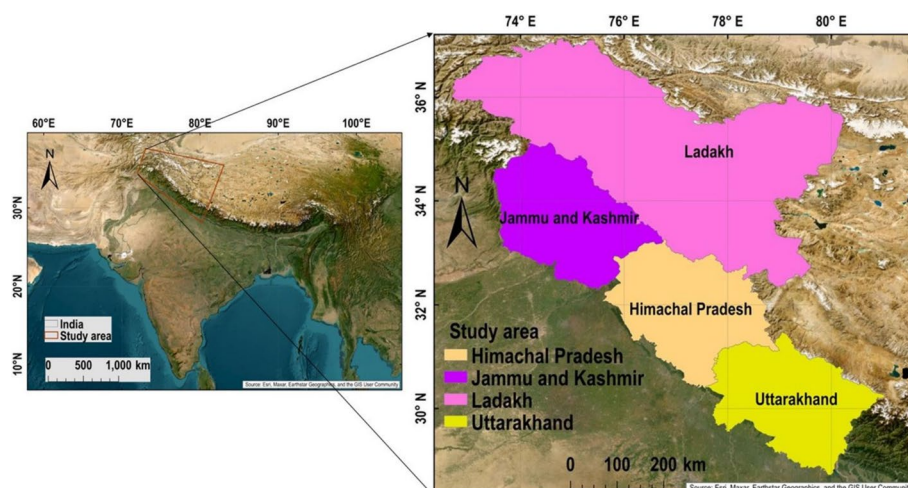


Fig. 1 Map of the Study Area with an Inset Map of India

situated at high altitudes and has a cold desert climate with harsh winters and cool summers [29]. Despite its arid climate, Ladakh receives occasional snowfall during winter. The total population of the entire study area is approximately 31 million people, comprising 11 million people from Uttarakhand, 7 million people from Himachal Pradesh, 13 million people from Jammu and Kashmir, and a population of 300,000 people from Ladakh, making the population of the entire study area around 31 million.

Urbanization in Uttarakhand, Himachal Pradesh, Jammu and Kashmir, and Ladakh presents a multifaceted picture influenced by geographical, cultural, and socio-economic factors unique to each region [40]. In Uttarakhand, urbanization is primarily concentrated in the plain areas of Dehradun, Haridwar, and Udham Singh Nagar, which are driven by proximity to transportation networks, government institutions, and educational facilities. In Himachal Pradesh, urbanization is characterized by the emergence of hill stations like Shimla, Manali, and Dharamshala as popular tourist destinations and administrative centers [40]. Jammu and Kashmir, with its capital city, Srinagar, experiences urbanization influenced by political unrest, tourism, and strategic importance. In Ladakh, urbanization is relatively slower compared to other regions due to its remote location and harsh climatic conditions. Towns like Leh and Kargil are witnessing gradual urban development driven by tourism, government initiatives, and increasing connectivity. Rapid urban growth in these areas has posed challenges in infrastructure development, environmental degradation, and rural-to-urban migration. However, unplanned urban expansion threatens the fragile Himalayan ecosystem and traditional livelihoods, especially in ecologically sensitive areas. The influx of tourists and changes in traditional lifestyles raise concerns about cultural preservation and sustainable development. Overall, urbanization in these Himalayan states reflects a complex interplay of economic aspirations, environmental concerns, and socio-cultural dynamics, necessitating holistic planning and governance strategies to ensure balanced and sustainable growth.

3 Data source and method

The details of the data used in the present study are given in Table 1.

The methodology of the present study comprises data acquisition, data processing, LULC classification using the Random Forest (RF) machine learning algorithm, and LULC prediction using MLP-ANN-MCM, as shown in Fig. 2.

3.1 Data acquisition

3.1.1 Sentinel-2 and its acquisition and processing

In the present study, high-resolution data of 10 m resolution of Sentinel-2 MSI (Level 1C) data, Earth observation satellites launched by the Copernicus program of the European Space Agency for the years 2016, 2019, and 2022, were used and processed in Google Earth Engine (GEE) [63]. It was launched in 2015 with a swath width of 290 km

Table 1 Details of the dataset used in the study

Data	Description	Source	Layer generated	Format
Digital Elevation Model (DEM)	SRTM 1 arc-second (30 m resolution)	USGS Earth Explorer (https://earthexplorer.usgs.gov/)	Elevation profile, slope, aspect, and stream distance	Raster (Tiff)
Harmonized Sentinel-2 MSI	(10 m resolution) For the years 2016, 2019 and 2022	European Union/ESA/Copernicus, Google Earth Engine (GEE)	LULC map	Raster (Tiff)

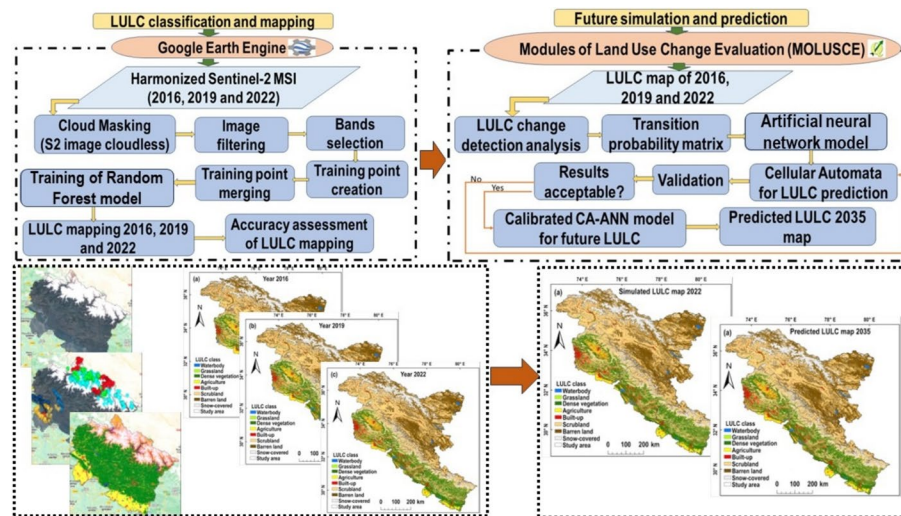


Fig. 2 Workflow of LULC Classification, Mapping, and Prediction

and is freely accessible. The wide swath and free data access help in the easy availability and processing of large areas. It consists of 13 spectral bands at different resolutions, i.e., four visible and near-infrared (NIR) bands (10 m), six red-edge and shortwave infrared (SWIR) bands (20 m), and three atmospheric correction bands (60 m) with a spectral range of 0.4 m to 2.4 m [73]. It has a temporal resolution of 5 days, and due to this high temporal resolution, the changes that occur within a short period can be monitored [103]. Due to low radiometric calibration uncertainty ranging from 0.03 to 0.4%, Sentinel-2 images have the potential to produce highly accurate and reliable results [81]. Different levels of Sentinel-2 data, such as Level-1B, Level-1C, and Level 2A, are produced by different processing stages [86].

The present study used Sentinel 2 Level 1C with geometric and radiometric corrections. It comprises $100 \times 100 \text{ km}^2$ tiles in an orthorectified format in UTM/WGS 84 projection. The orthorectified satellite image with the least cloud cover was imported into GEE, and the LULC map was prepared using the GEE algorithm. The GEE algorithm is created using a Google server and Javascript. After importing the image into GEE, cloud shadow and cover were removed by performing bitwise operations on the QA60 quality bands. The Harmonised Sentinel 2 MSI provides a time series of images, and in the present study, the number of images used in 2016, 2019, and 2022 is 1360, 2146, and 2712, respectively. The date range of the satellite image for each selected year ranges from 1st of January to 31st of December. For each year, a composite image was created using the median filter by assigning a median value to each pixel.

3.2 LULC mapping using machine learning (ML)-based random forest (RF) in GEE

The supervised LULC classification of the study area was performed for the years 2016, 2019, and 2022 using machine learning techniques of Random Forest (RF) in GEE. Machine learning-based accurate LULC classification requires adequate training samples, so a total of 3568, 3642, and 3547 pixels for the years 2016, 2019, and 2022, respectively, were extracted based on random distribution within LULC classes by a visual interpretation of the Google Earth high-resolution satellite imagery. There are eight LULC classes, including waterbody, built-up, agricultural land, grassland, dense

vegetation, scrubland, barren land, and snow-covered area. All the extracted samples were divided into training (80%) and testing (20%) datasets.

These training samples were then used to train the RF classifier, an ensemble learning method based on decision tree combinations. Breiman [21] introduced the RF classifier, the most commonly used machine learning technique that builds an ensemble classifier by combining several tree predictors [73]. RF classifiers in GEE generally use six input parameters, which are the number of trees used in classification, the number of variables used at each node, random seed variable for decision tree construction, minimum leaf population, bagged fraction of the input variables for each decision tree, and out-of-bag mode. The overall classification accuracy increases with the number of trees [21]. Variable importance (VI) is one of the RF classifier products that measures the impact of each variable on the overall performance of the model. It is capable of handling large and complex datasets while providing robust and accurate predictions, and due to its simplicity, efficiency, and ability to handle high-dimensional data with complex relationships, it is widely used for a wide range of applications in remote sensing, environmental monitoring, urban planning, and agriculture [77].

Random Forest creates an ensemble of decision trees, each trained on a random subset of the training data and features [35]. This randomness helps to introduce diversity among the trees, reducing the likelihood of overfitting and improving generalization. During the training process, the algorithm measures the importance of each feature in determining the class labels and allows to focus on the most informative features for classification, which is particularly useful in land use and land cover classification where certain features (e.g., vegetation indices, elevation, texture) might be more relevant than others [2]. It generally employs a technique known as bagging, where each decision tree is trained on a bootstrapped sample of the training data (sampling with replacement). This further enhances the diversity among the trees and helps to reduce variance, leading to a more stable and accurate model. In the classification phase, each decision tree independently predicts the class label for a given sample, and the final class label is determined through a voting mechanism, where the majority class predicted by the individual trees is chosen as the overall prediction [21]. This ensemble approach helps to mitigate the risk of bias and errors inherent in individual decision trees. In the present study, Number of Trees = 500; variablesPerSplit = 0; minLeafPopulation = 1; bagFraction was set to 0.5 are selected as important parameters.

3.3 Accuracy assessment of LULC map

The classified LULC map obtained by the RF classifier was validated using a confusion matrix, a built-in algorithm in GEE [123]. A confusion matrix, also known as an error matrix, is used to evaluate the performance of a classification model. It summarizes the predictions made by a classification model compared to the actual ground truth labels in a dataset. It is represented in a tabular form as a square matrix, where each row represents the actual class labels, and each column represents the predicted class labels [63]. The diagonal elements of the matrix represent the instances that were correctly classified, while the off-diagonal elements represent instances that were misclassified [39]. From the misclassified classes, the commission and omission errors were quantified via Producer Accuracy (PA) and User Accuracy (UA), respectively, as given in Eqs. (1) and (2).

$$PA = \frac{\text{Number of correctly classified pixels in each class}}{\text{Total number of reference pixels in that class (column sum)}} \times 100 \quad (1)$$

$$UA = \frac{\text{Number of correctly classified pixels in each class}}{\text{Total number of reference pixels in that class (row sum)}} \times 100 \quad (2)$$

Overall accuracy (OA) and Kappa coefficient (KC) were also calculated using 20% of the total datasets as testing data for the three years. Overall accuracy (OA), as defined by Eq. (3), is used to evaluate classification model performance and is expressed as the proportion of correctly classified sample pixels out of the total number of pixels in the dataset [73].

$$OA = \frac{\text{Number of correctly classified pixels diagonally}}{\text{Total number of reference pixels}} \times 100 \quad (3)$$

The Kappa coefficient (KC), as given in Eq. (4), also known as Cohen's Kappa statistic, is defined as the ratio of correctly classified sample points to the number of overall sample points [103]. A value close to 1 indicates the strength of agreement between the classifier's predictions and the true labels.

$$KC = \frac{N \sum_{i=1}^r a_{ii} - \sum_{i=1}^r (a_{i+})(a_{+i})}{N^2 - \sum_{i=1}^r (a_{i+})(a_{+i})} \quad (4)$$

where N is equal to the total number of pixels, r denotes the number of rows, a_{ii} number of observations in row i and column i , a_{+i} and a_{i+} are the marginal totals of row (r) and column (i), respectively. The population error matrix by Olofsson et al. [76] can also be used for the accuracy assessment of LULC maps. The methodology proposed by Wickham et al. [116] can be applied for the accuracy assessment of LULC changes.

F1-score is the most suitable metric for assessing class-level accuracy, and it is also calculated from the confusion matrix [40]. It is calculated by combining both metrics and also gives equal importance to both, Producer's accuracy as precision and User's accuracy as recall, as given in Eq. (5).

$$F1 - score = \frac{2 \times UA \times PA}{UA + PA} \quad (5)$$

3.4 LULC pattern analysis

In this study, the MOLUSCE plugin in QGIS 2.18.15 simulated LULC transitions among LULC classes and estimated spatiotemporal changes for 2016–2019, 2016–2022, and 2019–2022. The MOLUSCE plugin is an open-source model developed by Asia Air Survey QGIS 2.0 for the analysis, modelling and simulation of LULC changes. It consists of well-known algorithms and models namely artificial neural networks (ANNs), multi-criteria evaluation (MCE), weights of evidence (WoE), logistic regression (LR), and Monte Carlo cellular automata (CA) models. The model analyze the degree and spatiotemporal change of LULC and forecast the future LULC by utilizing socioeconomic and environmental parameters as predictors. Studies conducted by Alshari and Gawali [11], Das et al. [30], Kamaraj and Rangarajan [48], Muhammad et al. [69], Ahmad et al. [6], Amgoth

et al. [12], Değermenci [31], Lukas et al. [65], Bathe and Patil [17] and Raza et al. [88] found that the CA-ANN technique of MOLUSCE plugin for the modelling and simulation of future LULC is highly effective.

The prediction of future LULC using the plugin consists of several steps as follows: (i) analysis of LULC change, (ii) transition potential modeling using MLP-ANN-CA, (iii) prediction and validation of simulated LULC, and (iv) preparation of LULC prediction map for the year 2035. It applies the classic realization of Multilayer perceptron (MLP), where the input data is a collection of pixels of the initial state raster and factor raster layers, and the target output data is the change map [106]. For the generation of training data, MOLUSCE uses the assigned land use and cover maps. From these LULC maps of different years, transition maps are generated to show the changes between different classes over time. These transitions form the target outputs for the MLP model [37]. The corresponding explanatory variables for the same time periods are extracted and used as input features. For the training of the MLP-ANN-MCM model in MOLUSCE, prepared input features (explanatory variables) and the target outputs (land use changes) are used [13]. During training, the model learns to associate patterns in the input data with specific types of land use changes. This model uses historical land cover data and influencing factor raster layers (e.g., slope, elevation, distance to streams, aspect) to understand past transition trends [15]. Climate and urbanization rates are generally assumed to follow the same patterns as those observed during the baseline period [69].

3.4.1 Analysis of change in the area of LULC classes and transition potential modeling using multi-layer perceptron artificial neural network-Markov Chain model (MLP-ANN-MCM)

LULC change detection analysis for the years 2016–2019, 2016–2022, and 2019–2022 was carried out for the study region. Area changes and transition matrix were created for the three time periods. There are eight classes of LULC in the present study, so the transition matrix consists of 8 rows and 8 columns, giving 64 LULC class transitions. In the present study, the major LULC class transitions considered were (i) agriculture to built-up, barren land, dense vegetation, scrubland, and grassland, (ii) dense vegetation to agriculture, built-up, barren land, scrubland, and grassland, (iii) barren land to dense vegetation, agriculture, built-up, scrubland, and grassland.

The change analysis of LULC was further applied for transition potential modeling using the MLP-ANN-MCM approach. Certain spatial variables significantly influence the prediction of the LULC map. In the present study for the simulation of LULC using MOLUSCE, four additional raster layers, namely elevation profile, slope, aspect, and distance from the streams, were considered, as shown in Fig. 3a, b, c, and d. The raster layers were prepared using ArcMap, and Pearson's correlation was also calculated as the variables were continuous, as given in Table 2. The elevation profile, slope, aspect, and distance from streams were considered explanatory factors, as these layers are indications of physical and anthropogenic influences on LULC changes over a specific period [6].

The MLP-ANN-MCM, commonly utilized to model intricate nonlinear relationships between independent and dependent variables, operates on a supervised backpropagation algorithm within a feedforward artificial neural network structure [10]. It comprises three layers, i.e., input, hidden, and output layers, data distribution to neurons occurs via the input layer, while resultant data is obtained from the output layer [113].

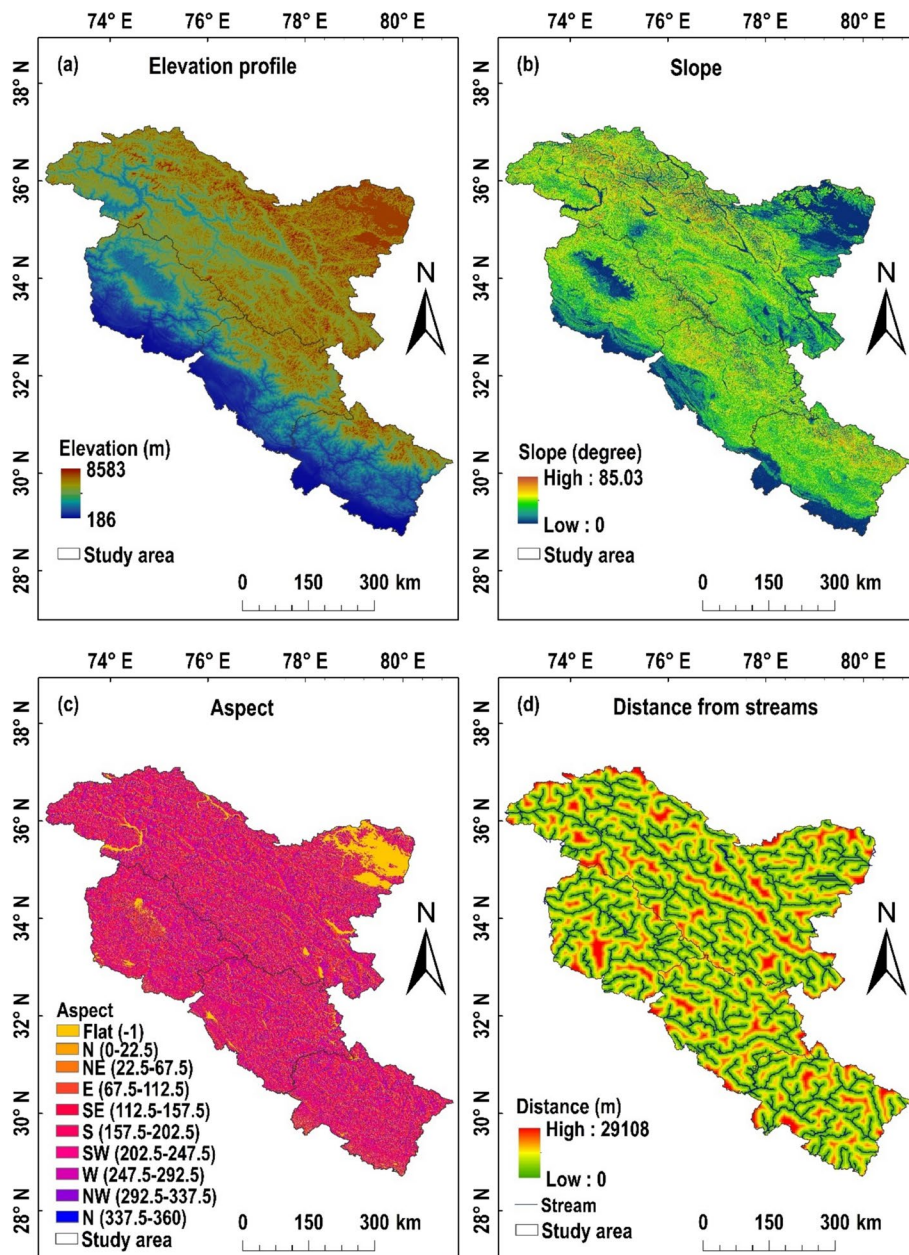


Fig. 3 Spatial variables analyzed in the study area for LULC prediction **a** Elevation, **b** Slope, **c** Aspect, **d** Distance from stream

Table 2 Pearson's correlation

Variables	Slope	Aspect	Distance from stream	Elevation
Slope	1	0.141	-0.032	0.046
Aspect		1	-0.064	-0.129
Distance from stream			1	0.238
Elevation				1

Positioned between the input and output layers, the hidden layer detects pattern alterations between input and output. Each neuron computes values based on the products of preceding-layer node values and network weights, and subsequently passes them through an activation function such as a sigmoid [115]. The backpropagation algorithm iteratively adjusts network weights to minimize errors between node outcomes and expected results. The derived transition potentials are further utilized to forecast future LULC classes.

3.4.2 Validation of simulated LULC

The present study used the MLP-ANN-MCM technique inside the MOLUSCE plugin to model the transition potentials and forecast future scenarios. According to Rahman et al. [85], Abbas et al. [1], Sarkar et al. [93], and Baig et al. [16], the CA-ANN approach is found to be more effective than linear regression. Based on the LULC maps of 2016 and 2019, the LULC map of 2022 was simulated using ANN approach, and the validation of the model and prediction accuracy were analyzed using the kappa validation technique and comparison of LULC maps. The parameters selected for the simulations are 1000 iterations, a neighborhood value of 1×1 pixel, a learning rate of 0.100, 100 hidden layers, and a momentum of 0.05. were selected. After analyzing the model accuracy by calculating percent correctness, kappa (overall), kappa (history), and kappa (loc), the LULC map of 2019 and 2022 for the study area was used to predict the LULC for 2035.

4 Results

4.1 LULC classification and its mapping using ML-based Random Forest (RF) in GEE

In the present study, an ML-based RF algorithm was applied to prepare the LULC map of the study area in GEE for 2016, 2019, and 2022, as shown in Fig. 4a, b, and c. The LULC classes were supported by Figs. 5. As suggested by the previous studies, the RF model is suitable for the classification of LULC due to its accuracy, efficiency, and suitability. The LULC of the study was divided into eight classes: waterbody, dense vegetation, grassland, agriculture, built-up, scrubland, barren land, and snow-covered area. The area distribution of the LULC classes for 2016, 2019, and 2022 is shown in Fig. 6a, b, and c, respectively. There was an increase in the scrubland from 31.56% to 35.35% and a decrease in barren land from 32.59% to 26.94% from the year 2016 to 2022, and the majority of the study area was dominated by scrubland and barren land. Dense vegetation slightly decreased from 17.77% in 2016 to 16.67% in 2022. The area occupied by built-up area was found to be 2.49%, 2.84%, and 3.66% for 2016, 2019, and 2022 respectively, showing an increasing trend. The area occupied by agricultural land was constant from 2016 to 2019, occupying 3.20% of the total study area, but in 2022, a slight decrease of 0.16% was noticed.

A similar trend was observed for waterbodies: from 2016 to 2019, the area covered by waterbodies remained constant at 0.85%, but in 2022 it decreased to 0.78%. From the region-wise LULC distribution, it was observed that Ladakh is the major contributor to the waterbodies, scrubland, barren land, and snow-covered areas. Ladakh contributed 46.68%, 46%, and 47.12% of the total waterbody of the study area in the years 2016, 2019, and 2022, respectively. In 2016, 2019, and 2022, respectively, about 46.49%, 50.66%, and 51.25% of the total scrubland was dominated by the Ladakh region, as shown in Fig. 7a, b, and c. For the total barren land in the study area in 2016, 2019, and 2022, the Ladakh

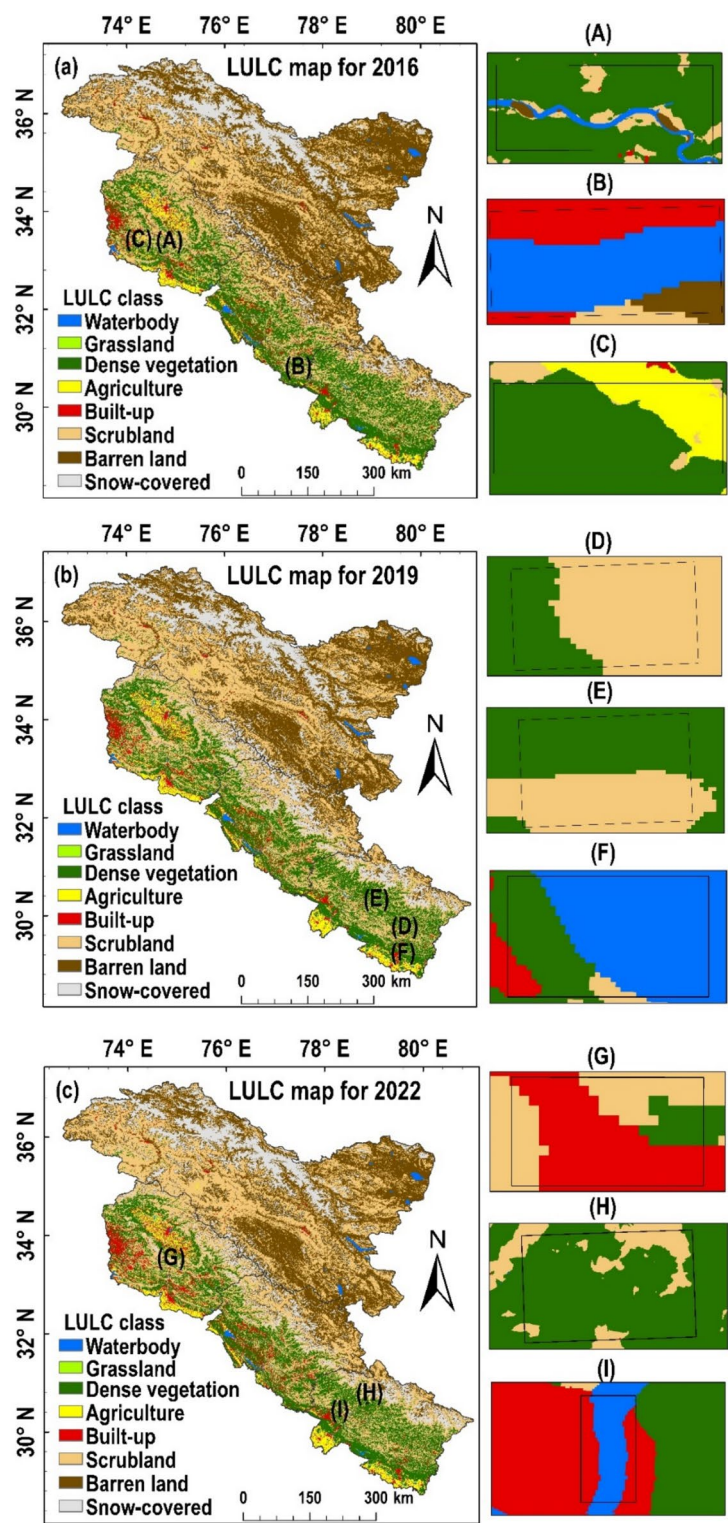


Fig. 4 Land Use and Land Cover (LULC) maps of the study area for three different years **a** 2016, **b** 2019, and **c** 2022



Fig. 5 Field photographs of Land Use and Land Cover (LULC) maps of the study area for three different years: 2016, 2019, and 2022

region contributed about 79.95%, 85.61%, and 83.17%, respectively. The majority of the total dense vegetation area lies in Uttarakhand. It was found that for 2016, 2019, and 2022, Uttarakhand contributed about 40.80%, 38.10%, and 40.66%, respectively, of the total dense vegetation of the study area. Most of the agricultural land, grassland, and built-up area were found to be dominated in the Jammu and Kashmir region.

In 2016, Jammu and Kashmir contributed 49.73% of the total agriculture of the study area, which then reduced to 47.92% in 2019 and 47.81% in 2022. Jammu and Kashmir comprised about 35.92% of the total grassland of the study area in 2016, 63.01% in 2019, and 50.99% in 2022. The trend in built-up shows that it was maximum for the Jammu and Kashmir region, with 49.94% of the total built-up of the study area. It shows an increasing trend for Jammu and Kashmir, 50.46% in 2019 and 51.67% in 2022. Following Ladakh, Himachal Pradesh accounts for the second major contributor of waterbodies, scrubland, barren land, and snow-covered areas of the total study area.

4.2 Accuracy assessment

In the present study, the accuracy of LULC maps for the years 2016, 2019, and 2022 prepared by the machine learning algorithm RF using the GEE was assessed using overall

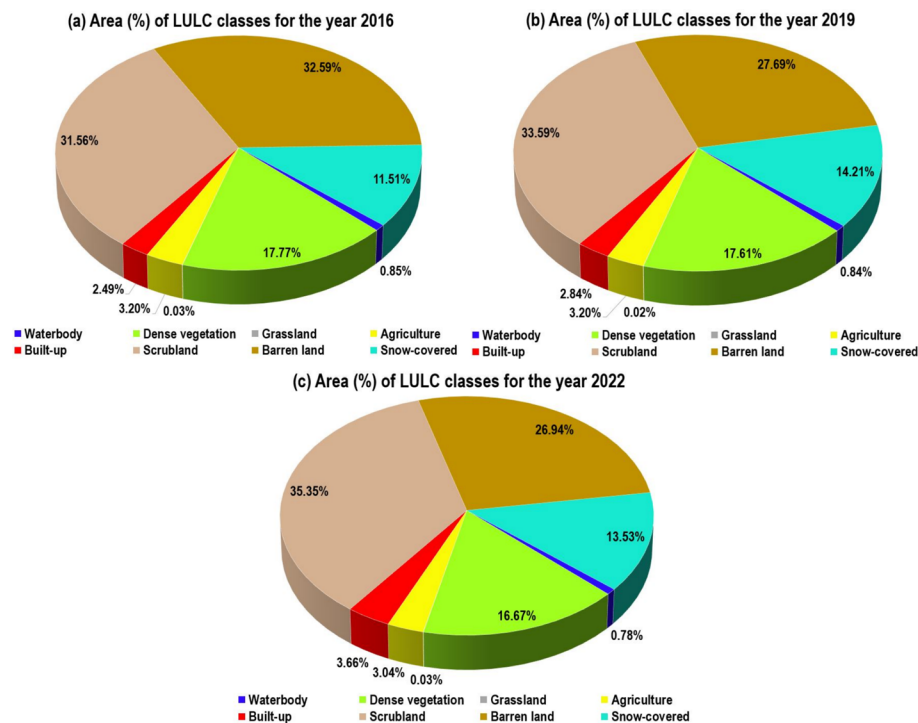


Fig. 6 Area distribution of different Land Use and Land Cover (LULC) classes for the year **a** 2016, **b** 2019, and **c** 2022

accuracy (*OA*), Kappa coefficient (*KC*), user's accuracy (*UA*) and producer's accuracy (*PA*) for each LULC class, as given in Table 3. The *OA* of the LULC maps for 2016, 2019, and 2022 was 88.76%, 87.64%, and 87.78%, respectively, indicating good performance of the RF model. The value of *KC* was also found to be close to 1, i.e., 0.872, 0.859, and 0.860 for the years 2016, 2019, and 2022, respectively. The *F1*-score was also greater than 80 for all LULC classes for the three years, as shown in Table 3. Among all LULC classes over the three years, the highest *OA* and *KC* values were observed in 2016. In 2016, *UA* values greater than 91% and *F1*-score exceeding 0.91 were observed for waterbody, dense vegetation, and barren land, highlighting the clear spectral separability. The LULC classes, such as grassland, snow-covered and scrubland, have comparatively lower *F1*-scores, ranging between 0.84 and 0.85, indicating confusion with adjacent vegetated classes due to spectral similarity. In 2019, *OA* slightly decreased to 87.64%, with *KC* equal to 0.859. Waterbody and dense vegetation perform well, with *F1*-scores greater than 0.91. Built-up and agricultural classes remained stable, with *F1*-scores around 0.86–0.89. In 2022, *OA* marginally improved to 87.78%, and *KC* = 0.860. Waterbodies, dense vegetation, agricultural land, and barren land performed well, with *F1*-Scores between 0.89 and 0.92. In all three years, grassland and scrubland have lower accuracy due to spectral overlap.

4.3 LULC change detection and transition analysis

In the present study, change detection of LULC classes was analyzed for the three time periods: period 1 (2016–2019), period 2 (2019–2022), and period 3 (2016–2022), as shown in Table 4. From 2016 to 2019, an increase of 0.35% was observed in the built-up areas, whereas LULC classes like waterbody, grassland, and barren land decreased by 0.01%, 0.01%, and 4.90%, respectively. For the agricultural land, no change in the area was observed during this period. For the second period (2019–2022), rapid growth in the

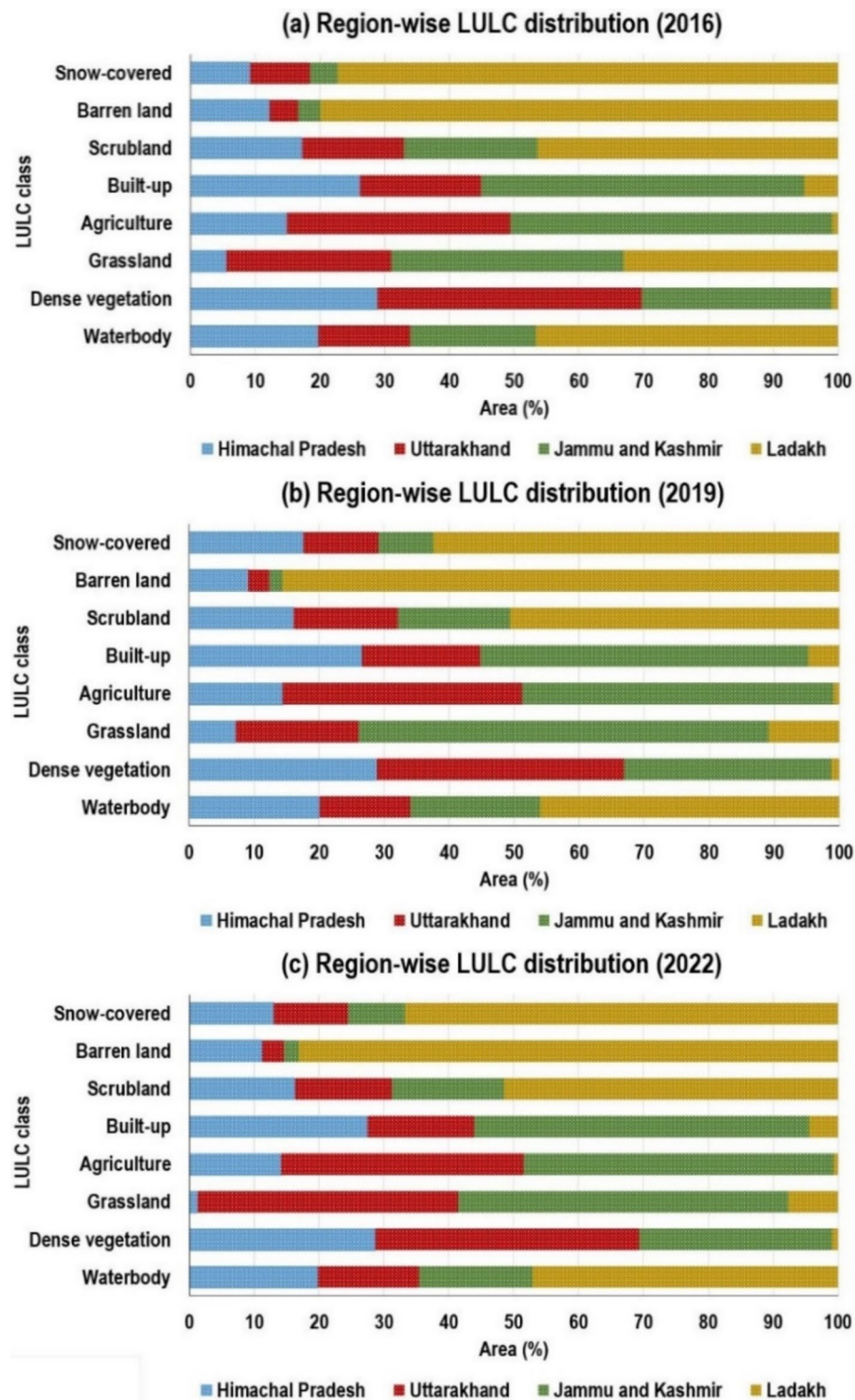


Fig. 7 Region-wise LULC distribution for the year **a** 2016, **b** 2019 and **c** 2022 across four regions: Himachal Pradesh, Uttarakhand, Jammu and Kashmir, and Ladakh

built-up areas by 0.82% increase was observed, and a decrease in agricultural land and dense vegetation by 0.16% occurred. During this period, there was a decrease in waterbody by 0.06%, which was higher than in period 1. For the second period, the grassland area shows no significant changes. From 2016 to 2022, the built-up area grew by 1.17%,

Table 3 Accuracy assessment of LULC for 2016, 2019 and 2022

Year	Class	WB	DV	GL	AG	BLT	SCL	BL	SCD	TU	UA (%)	F1-score
2016	WB	77	0	3	0	0	0	0	3	83	92.77	92.22
	DV	0	80	5	0	0	2	0	0	87	91.95	91.95
	GL	1	5	83	2	3	3	0	1	98	84.69	84.27
	AG	2	0	2	74	4	2	0	1	85	87.06	90.80
	BLT	0	2	3	2	86	3	0	3	99	86.87	86.43
	SCL	2	0	3	0	2	75	0	5	87	86.21	85.23
	BL	0	0	0	0	2	0	78	2	82	95.12	95.71
	SCD	2	0	0	0	3	4	3	81	93	87.10	85.71
	TP	84	87	99	78	100	89	81	96	714	OA = 88.76% KC = 0.872	
	PA (%)	91.67	91.95	83.84	94.87	86.00	84.27	96.30	84.38			
2019	WB	81	0	3	0	0	0	0	3	87	93.10	92.04
	DV	2	88	3	1	4	2	0	0	100	88.00	91.19
	GL	1	2	67	2	1	1	2	0	76	88.16	81.71
	AG	2	0	2	76	2	2	0	1	85	89.41	89.41
	BLT	0	2	3	2	80	3	0	3	93	86.02	86.02
	SCL	2	1	3	1	2	81	0	3	93	87.10	85.26
	BL	0	0	2	2	2	3	71	2	82	86.59	89.87
	SCD	1	0	5	1	2	5	3	87	104	83.65	85.71
	TP	89	93	88	85	93	97	76	99	720	OA = 87.64% KC = 0.859	
	PA (%)	91.01	94.62	76.14	89.41	86.02	83.51	93.42	87.88			

Table 3 (continued)

Year	Class	WB	DV	GL	AG	BLT	SCL	BL	SCD	TU	UA (%)	F1-score
2022	WB	83	1	2	0	0	1	0	4	91	91.21	92.22
	DV	1	92	1	1	4	2	0	0	101	91.09	91.54
	GL	1	1	56	2	1	1	2	0	64	87.50	81.16
	AG	2	0	2	79	2	2	0	1	88	89.77	89.77
	BLT	1	3	0	2	80	3	0	3	92	86.96	87.43
	SCL	0	0	5	1	2	63	0	3	74	85.14	81.82
	BL	0	1	1	2	2	3	84	2	95	88.42	91.30
	SCD	1	2	7	1	0	5	3	88	107	82.24	84.61
	TP	89	100	74	88	91	80	89	101	712	OA = 87.78%	
	PA (%)	93.26	92.00	75.68	89.77	87.91	78.75	94.38	87.13		KC = 0.860	

***Waterbody (WB), dense vegetation (DV), grassland (GL), agriculture (AG), Built-up (BLT), scrubland (SCL), barren land (BL), snow-covered (SCD), total producer (TP), Total user (TU), user's accuracy (UA), producer's accuracy (PA), overall accuracy (OA), Kappa coefficient (KC)

Table 4 Area distribution and change analysis of LULC classes

LULC class	2016		2019		2022		Changes in 2016–2019		Changes in 2019–2022		Changes in 2016–2022	
	Area (km ²)	Area (%)	Area (km ²)	Area (%)	Area (km ²)	Area (%)	Area (km ²)	Area (%)	Area (km ²)	Area (%)	Area (km ²)	Area (%)
Waterbody	2799.83	0.85	2759.58	0.84	2571.63	0.78	−40.25	−0.01	−187.96	−0.06	−228.21	−0.07
Dense vegetation	58,544.00	17.77	58,017.39	17.61	54,936.50	16.67	−526.61	−0.16	−3080.89	−0.93	−3607.50	−1.09
Grassland	103.68	0.03	80.22	0.02	87.90	0.03	−23.46	−0.01	7.67	0.00	−15.78	0.00
Agriculture	10,549.60	3.20	10,550.06	3.20	10,011.36	3.04	0.45	0.00	−538.69	−0.16	−538.24	−0.16
Built-up	8199.54	2.49	9363.35	2.84	12,063.32	3.66	1163.80	0.35	2699.97	0.82	3863.78	1.17
Scrubland	104,008.11	31.56	110,675.17	33.59	116,471.43	35.35	6667.07	2.02	5796.25	1.76	12,463.33	3.78
Barren land	107,395.07	32.59	91,236.67	27.69	88,775.13	26.94	−16,158.40	−4.90	−2461.54	−0.75	−18,619.94	−5.65
Snow-covered	37,911.33	11.51	46,828.72	14.21	44,593.90	13.53	8917.40	2.71	−2234.82	−0.68	6682.57	2.03

and this increase may be attributed to decreases in dense vegetation, grassland, and agricultural land. Figure 8a, b, and c represent the LULC change detection for the three time periods.

The transition probability matrix is given in Table 5, and according to the matrix for 2016–2019, waterbody, dense vegetation, agriculture, built-up, scrubland, barren land, and snow-covered have transition probability values greater than 0.80. Grassland decreased with the transition probability of 0.081 and contributed 0.078, 0.083, 0.081, 0.274, and 0.167 to a waterbody, dense vegetation, agriculture, scrubland, barren land, and snow-covered. During 2019–2022, the transition probabilities of waterbody, dense vegetation, agriculture, built-up, scrubland, barren land, and snow-covered were 0.854, 0.896, 0.852, 0.963, 0.896, and 0.816, respectively. The grassland contributed 0.089, 0.067, and 0.322 to the waterbody, dense vegetation, and scrubland, respectively, and dense vegetation contributed 0.082 to the scrubland. According to the transition potential matrix, from 2016 to 2022, the LULC classes such as waterbody, dense vegetation, agriculture, built-up, scrubland, and snow-covered area have probability transition values ranging from 0.812 to 0.960. Dense vegetation and barren land contributed 0.083 and 0.143 to the scrubland, respectively. The agricultural area contributed 0.095 to the built-up class, and 0.086 was contributed by dense vegetation to the waterbody.

4.4 LULC change transition potential modeling using multi-layer perceptron artificial neural network-Markov Chain model (MLP-ANN-MCM)

The LULC map for 2022 was simulated using the LULC transition from 2016 to 2019, based on MOLUSCE's approach, using MLP-ANN-MCM. A total of 5000 randomly distributed samples were used, and based on fixed values of the neighborhood (1×1 pixels), learning rate (0.01), maximum iteration (1000), hidden layer (10), and momentum (0.05), the LULC map for the year 2022 was simulated. The values of fixed overall accuracy, minimum error for validation, and validation kappa of transition potential modeling using MLP-ANN-MCM were 0.0016, 0.0284, and 0.8778, respectively. The simulated LULC map of 2022 was compared with the classified LULC map of the same year. The overall accuracy, validation kappa, and overall kappa were 85%, 0.882, and 0.831, respectively. The simulated LULC map for 2022 and the area of different LULC classes are given in Fig. 9a and b.

4.5 LULC prediction

After the validation of the model, future LULC classification and mapping for 2035 were predicted using the LULC data of 2019 and 2022, the spatial variables selected for simulation, the transition probability matrix, and a kappa value of 0.864 were obtained. The predicted LULC map for 2035 and the area distribution of different LULC classes are shown in Fig. 10a and b, respectively. The predicted LULC map for 2035 showed that from 2022 to 2035, there was an increase in agricultural and built-up areas of 1.20% and 2.31%, respectively, and a significant decrease in grassland of 2.59%, as indicated in Table 6.

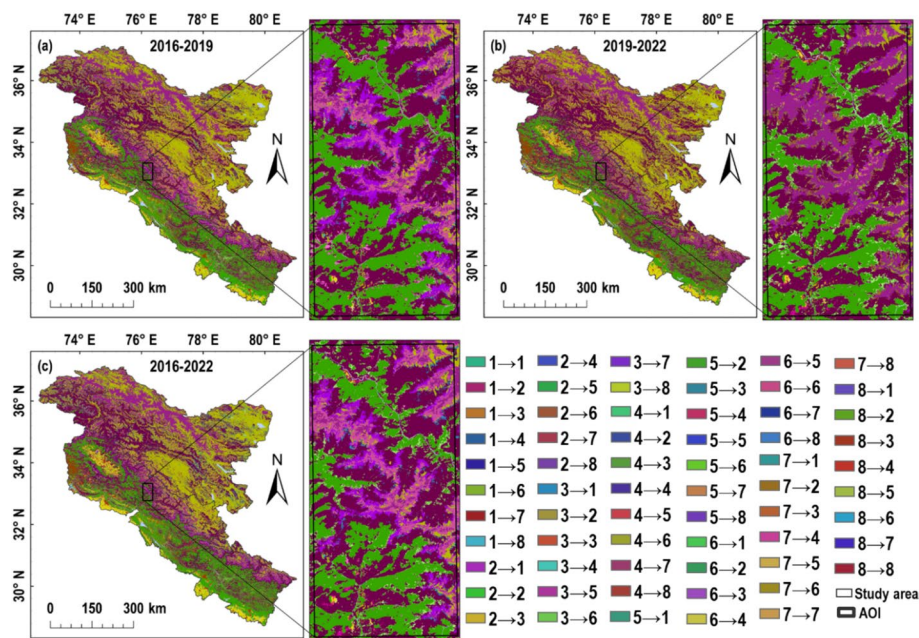


Fig. 8 Land use land cover transition maps **a** 2016–2019, **b** 2019–2022, and **c** 2016–2022 where (1) waterbody, (2) dense vegetation, (3) grassland, (4) agriculture, (5) built-up, (6) scrubland, (7) barren land, (8) snow-covered

5 Discussion

The IWH region is threatened by frequent natural hazards such as earthquakes, landslides, and floods. The frequency and intensity of these disasters are primarily governed by the LULC changes resulting from extensive development projects and rapid urban growth. Recent advancements in geospatial technology, particularly remote sensing data, have wide applications in LULC analysis and disaster monitoring and assessment [27, 39, 79]. In the present study, LULC classification and mapping for 2016, 2019, and 2022 were performed using Sentinel-2 image and RF algorithm in the GEE platform. The obtained LULC maps were further used to predict LULC using the MLP-ANN-MCM approach. The GEE platform utilized in the present study has proven highly effective for accessing and processing satellite imagery, similar to the studies conducted by Nasiri et al. [73] and Feizizadeh et al. [35]. The results show that GEE enables users to visualize, analyze, and interpret satellite data for diverse environmental applications. It rationalizes the classification process for extensive study areas, offering enhanced flexibility in performing image pre-processing tasks and accelerating and simplifying large-scale land-use mapping, aided by the robust computational capabilities of Google servers [25, 77, 103]. The Sentinel-2 image with 10 m resolution utilized in the present study is found to be highly suitable for integrating various LULC classifications with more accuracy for diverse LULC patterns similar to the studies of Colkesen and Kavzoglu [26], Varade et al. [108] and Chen et al. [24]. Integrating Sentinel-2 imagery with the RF classifier in GEE improves classification accuracy and efficiency. The present study used the RF algorithm for LULC classification and mapping, and it emerged as the best-performing model based on accuracy and kappa coefficient across all three years.

The LULC of the study area was classified into eight distinct classes: waterbody, dense vegetation, grassland, agriculture, built-up, scrubland, barren land, and snow-covered area. The analysis reveals that from 2016 to 2022, prominent trends in land

Table 5 Probabilities of transition between different LULC classes

LULC transition matrix									
	LULC class	WB	DV	GL	AG	BLT	SCL	BL	SCD
2019									
2016	WB	0.891	0.004	0.003	0.012	0.002	0.029	0.043	0.016
	DV	0.000	0.909	0.000	0.010	0.007	0.072	0.000	0.001
	GL	0.078	0.083	0.284	0.081	0.002	0.274	0.031	0.167
	AG	0.002	0.023	0.000	0.864	0.042	0.069	0.000	0.000
	BLT	0.001	0.019	0.000	0.018	0.926	0.034	0.002	0.000
	SCL	0.001	0.042	0.000	0.006	0.008	0.909	0.002	0.032
	BL	0.001	0.000	0.000	0.000	0.000	0.097	0.808	0.093
	SCD	0.000	0.000	0.000	0.000	0.000	0.011	0.109	0.880
2022									
2019	WB	0.854	0.007	0.005	0.017	0.005	0.049	0.053	0.010
	DV	0.000	0.896	0.000	0.005	0.015	0.082	0.000	0.001
	GL	0.089	0.067	0.469	0.025	0.002	0.322	0.004	0.021
	AG	0.001	0.023	0.001	0.852	0.064	0.057	0.000	0.000
	BLT	0.000	0.009	0.000	0.010	0.963	0.016	0.002	0.000
	SCL	0.001	0.023	0.000	0.005	0.013	0.938	0.002	0.017
	BL	0.001	0.000	0.000	0.000	0.000	0.055	0.896	0.048
	SCD	0.000	0.000	0.000	0.000	0.000	0.042	0.142	0.816
2022									
2016	WB	0.823	0.007	0.004	0.025	0.005	0.052	0.065	0.019
	DV	0.000	0.887	0.000	0.010	0.019	0.083	0.000	0.001
	GL	0.086	0.072	0.352	0.032	0.003	0.265	0.030	0.159
	AG	0.001	0.019	0.001	0.812	0.095	0.071	0.000	0.000
	BLT	0.001	0.009	0.000	0.011	0.960	0.017	0.002	0.000
	SCL	0.001	0.026	0.000	0.007	0.020	0.913	0.004	0.031
	BL	0.001	0.000	0.000	0.000	0.000	0.143	0.786	0.070
	SCD	0.000	0.000	0.000	0.000	0.000	0.009	0.100	0.891

Waterbody (WB), dense vegetation (DV), grassland (GL), agriculture (AG). Built-up (BLT), scrubland (SCL), barren land (BL), snow-covered (SCD)

cover dynamics occurred. The increase in scrubland and decrease in barren land suggests potential transformations within the landscape, influenced by climate change, anthropogenic activities, or natural processes. The slight decline observed in dense vegetation indicates potential environmental stressors affecting vegetative cover, and it also emphasizes the need for its conservation and restoration. The increase in built-up area from 2016 to 2022 highlights urbanization or infrastructure development, reflecting socio-economic dynamics shaping the study region. The changes in LULC over the studied period signify socio-economic transformations and necessitate sustainable land management strategies for mitigating environmental impacts and ensuring urban resilience. Region-wise LULC analysis revealed spatial heterogeneity in LULC distributions. Ladakh emerged as a significant contributor to waterbodies, scrubland, barren land, and snow-covered areas, dense vegetation is concentrated in Uttarakhand, and agricultural land, grassland, and built-up areas are predominant in the Jammu and Kashmir region. The result highlights the dominance of a certain land cover class in a specific region that geographical, climatic, and anthropogenic factors may influence.

The changes in LULC pattern were also analyzed for three time periods: 2016–2019, 2019–2022, and 2016–2022. According to the results, the observed transition highlights the evolving landscape patterns of the study region and underlying drivers of change. During the first period (2016–2019), an increase in built-up areas was accompanied by

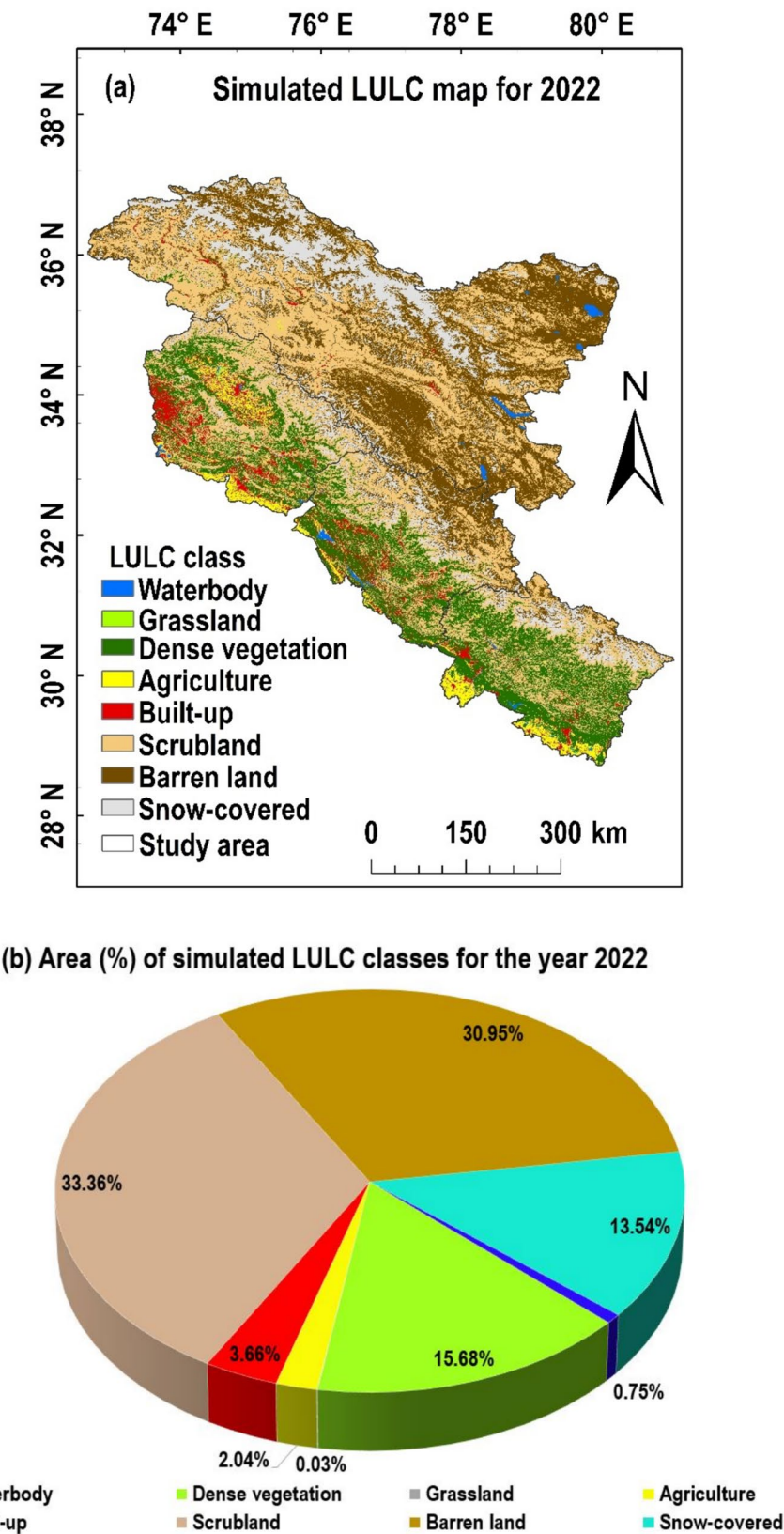


Fig. 9 a Simulated Land Use and Land Cover (LULC) map for 2022, b Area distribution of LULC class

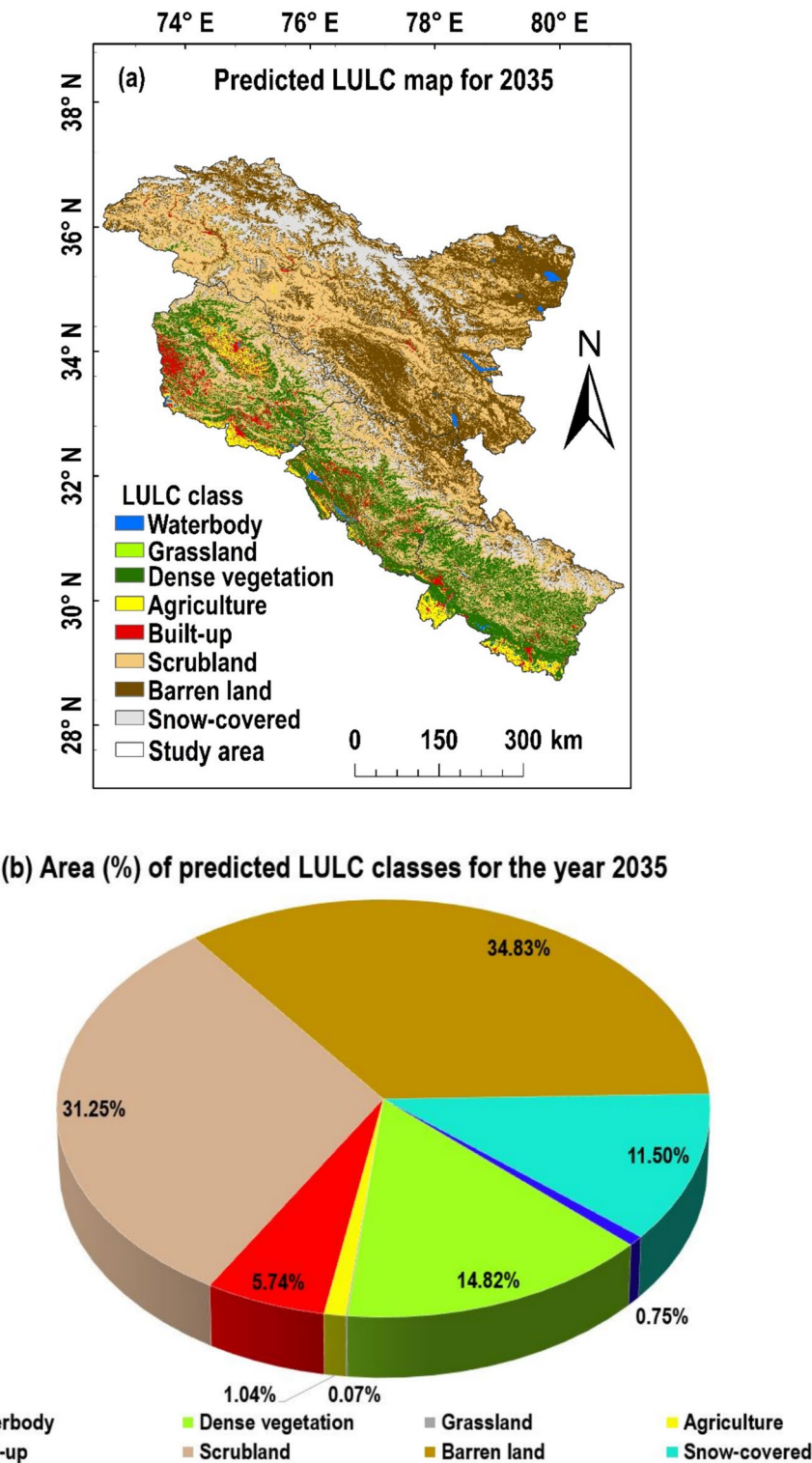


Fig. 10 a Predicted LULC map for 2035, b Area distribution of LULC class

declines in waterbodies, grasslands, and barren lands, and stability of agricultural land was observed. An increase in the built-up area and a decrease in natural surroundings indicate a shift towards urbanization and infrastructure development at the expense of natural habitats and ecosystem services. From 2019 to 2022 and 2016 to 2022, a

Table 6 Changes in LULC class from 2022–2035

LULC class	Changes in area (%) 2022–2035
Waterbody	– 1.04
Dense vegetation	– 0.02
Grassland	– 2.59
Agriculture	1.20
Built-up	2.31
Scrubland	– 0.35
Barren land	– 0.58
Snow-covered	1.27

substantial increase in built-up areas with declines in agricultural land and dense vegetation was observed, indicating an acceleration of significant urban growth and continued land conversion for urban development or other non-agricultural purposes. There is a decrease in grassland from 2016 to 2019 but from 2019 to 2022, an increase in grassland was observed. This trend may be attributed to various factors such as interannual climatic variability, particularly changes in precipitation influencing vegetation regeneration, minor spectral confusion between grassland and scrubland classes, and variations in grazing intensity or local land management practices.

The transition probability matrix helps to understand the dynamics of LULC change, highlighting the likelihood of transitions between different land cover classes over time. In the present study, the built-up area shows high transition probabilities, signifying persistent urbanization trends. At the same time, the contributions of dense vegetation, grasslands, and agricultural lands to other LULC classes emphasize the interconnectivity and complexity of landscape dynamics. The LULC transition over the studied period indicates the escalating human footprint and highlights the need for sustainable land use planning and development to mitigate adverse environmental impacts. The simulated LULC map 2022 was compared with the observed LULC map 2022 in the MOLUSCE plugin, and kappa statistics were calculated. The resulting Kappa coefficients were above the acceptable criteria of 80%, indicating that the accuracy was adequate for predicting future LULC [8, 48, 55, 90]. The kappa values found the model suitable for predicting LULC for 2035. The predicted LULC map for 2035 shows an increase in agricultural and built-up areas and a significant decrease in grassland.

The overall methodology and approach adopted in the present study were effective and reliable. It successfully addressed and identified the drivers of LULC change, such as urbanization, agricultural expansion, and ecosystem degradation. The findings of the present study help in understanding the patterns and processes of LULC change over the studied period. It also emphasizes the need for effective land use planning and management strategies to promote sustainable development and resilience in the face of environmental uncertainties.

Indian Western Himalayan region is facing continuous environmental challenges impacting the livelihoods of millions. The present scenario demands the development of policies promoting sustainable land use practices for the well-being of both present and future generations. According to the findings of the present study, the preservation of the diverse ecosystems of the study region is of utmost importance for maintaining the ecological balance. The decrease in agricultural land in Himachal Pradesh and Jammu & Kashmir regions could impact food security, connect with SDG 2 (Zero Hunger). Sustainable agriculture practices

and land management are essential to ensure food production while minimizing negative environmental impacts. The overall trend of increasing built-up areas and decreasing vegetation and agricultural land may contribute to climate change. The observed changes in snow-covered areas in Himachal Pradesh and Ladakh regions can relate to SDG 13 (Climate Action) emphasizing the need to take urgent action to combat climate change and its impacts, which includes sustainable land-use practices to reduce emissions and enhance resilience. The decline in dense vegetation areas raises concerns about biodiversity loss and ecosystem health, linking to SDG 15 (Life on Land). Sustainable land-use practices are crucial to preserving ecosystems, and efforts should be made to balance urban development with conservation to meet this goal. Policymakers can prioritize sustainable agricultural practices by promoting crop diversification and efficient water management techniques to enhance food security and minimize negative environmental impacts such as soil degradation, deforestation, and water scarcity. Policy development should consider climate change in land use planning and management strategies and mitigate its impact by promoting afforestation, reforestation, and sustainable land management. The community should engage and participate in considering local knowledge, priorities, and needs in decision-making processes related to land use planning and management. Different sectors, such as government agencies, civil society organizations, research institutions, and local communities, should be involved and collaborate by establishing multi-stakeholder platforms, sharing data and information, and fostering knowledge exchange and innovation to address challenges related to land use.

5.1 LULC Change hotspots vis-à-vis national programs and SDGs

Rapidly growing major LULC change hotspots in the IWH region, including Himachal Pradesh, Jammu & Kashmir, Ladakh (UT India), and Uttarakhand, are easily identified and are being monitored by the government space applications agencies of ISRO [75] and many other organisations [100]. The spotted consistent changes, from 2016 to 2019, reflect growth in built-up areas due to fast-paced urbanization and infrastructure development [100]. The simultaneous increase in developed/built-up land and decline in dense vegetation and agricultural areas hints towards a significant conflict between urban development and environmental conservation, raising concerns for ecosystem health and agricultural sustainability [87, 95, 112]. Notably, the highest deforestation rate has been observed between 2019 and 2022, with overall forest loss continuing, while new forest areas have diminished and extinct forest zones have increased, highlighting persistent challenges in forest management. Similarly, the decrease in water resources and waterbodies in Jammu & Kashmir underscores the urgent need for robust water resource management, and drastic grassland changes in Jammu & Kashmir, Himachal Pradesh, and Ladakh present risks for biodiversity and ecosystem stability [20]. Ladakh's dominance in scrubland and barren terrains is directly related to its harsh topography, while Uttarakhand remains notable for maintaining dense forest cover, and "forest-to-forest" transitions consistently cover the largest area, suggesting stability in some forested zones [49].

Addressing these changes, national government programs suggest direct and quantitative impacts aligned with Sustainable Development Goals (SDGs) 2 (Zero Hunger), 6 (Clean Water), 13 (Climate Action), and 15 (Life on Land). For example, the Pradhan Mantri Krishi Sinchai Yojana [83] promoted micro-irrigation, country-wide, provides immense potential for enhancing agricultural productivity in the recently transitioned forest-to-agricultural

lands, set up in hilly terrains of the IWH region. Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) scheme [66] will provide enough person-days of employment for the poor sections of the population in the region, with extensive watershed and soil conservation works supporting ecosystem services and resilience. The Jal Jeevan Mission (JJM, 2025) connected over 15.72 crore rural households, throughout India, to safe drinking water by October 2025, boosting SDG-6 and resource stability and monitoring of waterbodies in any form, rivers, ponds, glaciers [109], will hold very high potential for boosting and uplifting peoples' lives. PMFBY provided crop insurance for about 40% of India's farmers, facilitating risk transfer and resilience [82], and NMSA prioritized soil–water conservation across vulnerable regions [74]. National SDG indices reflect progress, with India's composite SDG score improving from 66 (2020–21) to 75 (2023–24), and the number of 'Front Runner' states in SDGs increasing from 17 to 32, underscoring substantial policy-driven improvements in land, water, and climate-related targets despite ongoing pressures from LULC transitions. These government schemes aim to benefit all citizens of Bharat. This is a work in progress, and the schemes have the potential to extend to the IWH region. The findings of this study can help ministries obtain the necessary information regarding changes in land categories and allocate resources from various schemes—such as JJM, MNREGA, PMKSY, PMFBY, and NSMA—more effectively to all citizens.

6 Conclusion

The present study assesses LULC in the Indian Western Himalayan region using Sentinel-2 imagery and an ML-based RF classifier in Google Earth Engine. Prediction of future LULC for 2035 was performed using the MLP-ANN-MCM approach. The findings indicate that climate change, human activities, and natural phenomena highly influence the LULC distribution. The transition probability matrix shows high transition probabilities for built-up areas, indicating ongoing urbanization trends. The predicted LULC map for 2035 indicates an increase in built-up and agricultural areas. The projected change in built-up area and agricultural land by the projected period (2035) has direct implications for SDG 2 (Zero Hunger) and SDG 15 (Life on Land), concerning sustainable management and preservation of productive ecosystems that are essentially required for the maintenance of food security and ecosystem services. The methodology employed in the present study can be applied to other regions facing rapid urbanization and LULC changes. The derived spectral-temporal metrics can be harnessed to create precise land use and land cover (LULC) maps utilizing the GEE cloud computing platform. These findings underscore the substantial contributions of cloud-based geocomputation platforms like GEE and modern Earth observation satellites such as Sentinel-2 in enhancing the accuracy of LULC mapping, monitoring, and modeling. It is also evident from the study that machine learning algorithms like MOLUSCE can efficiently analyze multispectral satellite imagery and offer highly accurate future land management simulations, with broad utility for fast-paced urbanizing regions, having alignment with SDG 13 (Climate action) and SDG 6 (Clean Water and Sanitation) as pinpointed highly accurate LULC monitoring helps in framing targeted adaptation and mitigation strategies intending to control land degradation, water resource safety policies, and enabling climate resilience. Paralleling acknowledging the study's limitations such as complexity in landscape mosaic, classification challenges, ground truthing constraints due to highly inaccessibility of the terrain, etc., the findings of the study lay high stress on the need for integrated and sustainable landuse planning leading to addressal of the drivers of LULC change, namely urbanization,

agricultural expansion, and ecosystem degradation, by considering the ecological, social, and economic dimensions. The present LULC prediction study can be extended by coupling with process-based models in the field of hydrology, climate, and ecology to evaluate impacts on floods, urban heat, carbon dynamics, and biodiversity. Future emphasis can also be given on the high-resolution and sub-pixel mapping, uncertainty quantification in forecasts, integration with climate change projections, and participatory mapping using citizen-generated data. Such planning must balance socioeconomic growth with environmental stewardship, directly supporting progress toward the 2030 SDG agenda through informed policy and practical tools for stakeholders addressing the intertwined drivers of land change—urbanization, agricultural expansion, and ecosystem vulnerability.

Acknowledgements

Authors are thankful to Shiv Nadar Institution of Eminence for help, funding and support.

Author contributions

Laxmi Gupta (L.G.): Conceptualization, Methodology, Visualization, Image Analysis, Investigation, Formal Analysis, Field Survey, Validation, Resources, Data Curation, Writing—Original Draft. Jagabandhu Dixit (J.D.): Conceptualization, Supervision, Project Administration, Methodology, Data Curation, Investigation, Formal Analysis, Visualization, Writing—Review & Editing. Prem Chandra Pandey (P.C.P.): Conceptualization, Methodology, Supervision, Project Administration, data analysis, Writing—Review & Editing, Final Review. Manish Pandey (M.K.): Conceptualization, Methodology, Visualization, Writing—Review & Editing, Final Review. All authors have read and approved the final manuscript.

Funding

Open access funding provided by Shiv Nadar University. This study is funded by the FGIR/2023/Proposal/27 through Faculty Grant for Interdisciplinary Research (FGIR), Shiv Nadar Institution of Eminence and Shiv Nadar Foundation.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate

Not applicable. As study doesn't involve human participants, data, or tissues, meaning no formal ethical consent was needed.

Consent for publication

Not applicable. As study doesn't involve human participants, data, or tissues, meaning no formal ethical consent was needed.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Energy and Environment, Thapar Institute of Engineering and Technology, Patiala 147004, India

²Department of Civil Engineering, Shiv Nadar Institution of Eminence (Deemed to be University), Greater Noida, Uttar Pradesh 203207, India

³Department of Life Sciences, Shiv Nadar Institution of Eminence (Deemed to be University), Greater Noida, Uttar Pradesh 203207, India

⁴Marwadi University Research Center (MURC), Marwadi University, Rajkot, Gujarat 360003, India

⁵Department of Civil Engineering, Faculty of Engineering & Technology, Marwadi University, Rajkot, Gujarat 360003, India

Received: 4 August 2025 / Accepted: 18 February 2026

Published online: 22 March 2026

References

1. Abbas Z, Yang G, Zhong Y, Zhao Y. Spatiotemporal change analysis and future scenario of LULC using the CA-ANN approach: a case study of the Greater Bay Area, China. *Land*. 2021;10(6):584. <https://doi.org/10.3390/land10060584>.
2. Abijith D, Saravanan S. Assessment of land use and land cover change detection and prediction using remote sensing and CA Markov in the northern coastal districts of Tamil Nadu, India. *Environ Sci Pollut Res*. 2022;29(57):86055–67. <https://doi.org/10.1007/s11356-021-15782-6>.
3. Agrawal N, Dixit J. Assessment of landslide susceptibility for Meghalaya (India) using bivariate (frequency ratio and Shannon entropy) and multi-criteria decision analysis (AHP and Fuzzy-AHP) models. *All Earth*. 2022;34(1):179–201. <https://doi.org/10.1080/27669645.2022.2101256>.

4. Ahmad F, Goparaju L. Climate change and its impact on forest fire in the state of Himachal Pradesh and Uttarakhand states of India: remote sensing and GIS analysis. *Contemp Trends Geosci*. 2018;7(2):229–46. <https://doi.org/10.2478/ctg-2018-0016>.
5. Ahmad F, Goparaju L, Qayum A. LULC analysis of urban spaces using Markov chain predictive model at Ranchi in India. *Spat Inf Res*. 2017;25(3):351–9. <https://doi.org/10.1007/s41324-017-0102-x>.
6. Ahmad MN, Shao Z, Javed A. Modelling land use/land cover (LULC) change dynamics, future prospects, and its environmental impacts based on geospatial data models and remote sensing data. *Environ Sci Pollut Res*. 2023;30(12):32985–3001. <https://doi.org/10.1007/s11356-022-24442-2>.
7. Al Rifat SA, Liu W. Predicting future urban growth scenarios and potential urban flood exposure using Artificial Neural Network-Markov Chain model in Miami Metropolitan Area. *Land Use Policy*. 2022;114:105994. <https://doi.org/10.1016/j.landusepol.2022.105994>.
8. Alam A, Bhat MS, Maheen M. Using Landsat satellite data for assessing the land use and land cover change in Kashmir Valley. *GeoJournal*. 2020;85:1529–43. <https://doi.org/10.1007/s10708-019-10037-x>.
9. Aldiansyah S, Saputra RA. Comparison of machine learning algorithms for land use and land cover analysis using Google Earth Engine (case study: Wanggu watershed). *Int J Remote Sens Earth Sci (IJReSES)*. 2023;19(2):197–210. <https://doi.org/10.30536/ijreses.2022.v19.a3803>.
10. Al-Dousari AE, Mishra A, Singh S. Land use land cover change detection and urban sprawl prediction for Kuwait metropolitan region, using multi-layer perceptron neural networks (MLPNN). *Egypt J Remote Sens Space Sci*. 2023;26(2):381–92. <https://doi.org/10.1016/j.ejrs.2023.05.003>.
11. Alshari EA, Gawali BW. Modeling land use change in Sana'a city of Yemen with MOLUSCE. *J Sensors*. 2022;2022(1):7419031. <https://doi.org/10.1155/2022/7419031>.
12. Amgoth A, Rani HP, Jayakumar KV. Exploring LULC changes in Pakhal Lake area, Telangana, India using QGIS MOLUSCE plugin. *Spat Inf Res*. 2023;31(4):429–38. <https://doi.org/10.1007/s41324-023-00509-1>.
13. Anand B, Rekha RS, Radhakrishnan N, Ramaswamy K. Analysis of LULC change dynamics and its impact assessment using CA-ANN model in part of Coimbatore region, India. *GeoJournal*. 2023;88(6):5825–45. <https://doi.org/10.1007/s10708-023-10944-0>.
14. Arifeen HM, Phoungthong K, Mostafaeipour A, Yuanyai N, Yuanyai C, Techato K, et al. Determine the land-use land-cover changes, urban expansion and their driving factors for sustainable development in Gazipur Bangladesh. *Atmosphere Basel*. 2021;12(10):1353. <https://doi.org/10.3390/atmos12101353>.
15. Baghel S, Kothari MK, Tripathi MP, Singh PK, Bhakar SR, Dave V, et al. Spatiotemporal LULC change detection and future prediction for the Mand catchment using MOLUSCE tool. *Environ Earth Sci*. 2024;83(2):66. <https://doi.org/10.1007/s12665-023-11381-5>.
16. Baig MF, Mustafa MRU, Baig I, Takaijudin HB, Zeshan MT. Assessment of land use land cover changes and future predictions using CA-ANN simulation for Selangor, Malaysia. *Water Basel*. 2022;14(3):402. <https://doi.org/10.3390/w14030402>.
17. Bathe KD, Patil NS. Assessment of land use-land cover dynamics and its future projection through Google Earth Engine, machine learning and QGIS-MOLUSCE: a case study in Jagatsinghpur district, Odisha, India. *J Earth Syst Sci*. 2024;133(2):111. <https://doi.org/10.1007/s12040-024-02305-3>.
18. Beroho M, Briak H, Cherif EK, Boulahfa I, Ouallali A, Mrabet R, et al. Future scenarios of land use/land cover (LULC) based on a CA-Markov simulation model: the case of a Mediterranean watershed in Morocco. *Remote Sens*. 2023;15(4):1162. <https://doi.org/10.3390/rs15041162>.
19. Bhasin A, Dolker P, Raina P, Ghosal S. Land use and land cover change detection using remote sensing in the trans Himalayan region of Ladakh, India. *ECS Trans*. 2022;107(1):2985.
20. Bhat W. Kashmir's Drying Springs: A Silent Himalayan Water Crisis [WWW Document]. URL <https://www.indiawaterportal.org/groundwater/once-a-lifeline-now-a-memory-what-happened-to-kashmirs-water-springs> (2025). Accessed 11.20.25.
21. Breiman L. Random forests. *Mach Learn*. 2001;45:5–32. <https://doi.org/10.1023/A:1010933404324>.
22. Carrillo-Niquete GA, Andrade JL, Valdez-Lazalde JR, Reyes-García C, Hernández-Stefanoni JL. Characterizing spatial and temporal deforestation and its effects on surface urban heat islands in a tropical city using Landsat time series. *Landsc Urban Plan*. 2022;217:104280. <https://doi.org/10.1016/j.landurbplan.2021.104280>.
23. Cengiz AVCI, Budak M, Yağmur N, Balçık F. Comparison between Random Forest and support vector machine algorithms for LULC classification. *Int J Eng Geosci*. 2023;8(1):1–10. <https://doi.org/10.26833/ijeg.987605>.
24. Chen R, Yang H, Yang G, Liu Y, Zhang C, Long H, et al. Land-use mapping with multi-temporal Sentinel images based on Google Earth Engine in Southern Xinjiang Uygur Autonomous Region, China. *Remote Sens*. 2023;15(16):3958. <https://doi.org/10.3390/rs15163958>.
25. Chen S, Woodcock CE, Bullock EL, Arévalo P, Torchinava P, Peng S, et al. Monitoring temperate forest degradation on Google Earth Engine using Landsat time series analysis. *Remote Sens Environ*. 2021;265:112648. <https://doi.org/10.1016/j.rse.2021.112648>.
26. Colkesen I, Kavzoglu T. Ensemble-based canonical correlation forest (CCF) for land use and land cover classification using Sentinel-2 and Landsat OLI imagery. *Remote Sens Lett*. 2017;8(11):1082–91. <https://doi.org/10.1080/2150704X.2017.1354262>.
27. Costache R, Arabameri A, Costache I, Crăciun A, Islam ARMT, Abba SI, et al. Flood hazard potential evaluation using decision tree state-of-the-art models. *Risk Anal*. 2024;44:439–58. <https://doi.org/10.1111/risa.14179>.
28. Dad JM, Muslim M, Rashid I, Reshi ZA. Time series analysis of climate variability and trends in Kashmir Himalaya. *Ecol Indic*. 2021;126:107690. <https://doi.org/10.1016/j.ecolind.2021.107690>.
29. Dame J, Morup T. Environmental change and development in Ladakh, Indian Trans-Himalaya. B. Humbert-Droz (Ed.). Springer Nature; 2023. <https://doi.org/10.1007/978-3-031-42494-6>.
30. Das M, Mandal A, Das A, Pereira P. Land use and land cover change future projection in Kolkata Metropolitan Area, Eastern India. In: Mapping and forecasting land use. Elsevier; 2022. pp. 299–320. <https://doi.org/10.1016/B978-0-323-90947-1.00011-9>.
31. Değermenci AS. Spatio-temporal change analysis and prediction of land use and land cover changes using CA-ANN model. *Environ Monit Assess*. 2023;195(10):1229. <https://doi.org/10.1007/s10661-023-11848-9>.
32. Dhanaraj K, Angadi DP. Land use land cover mapping and monitoring urban growth using remote sensing and GIS techniques in Mangaluru, India. *GeoJournal*. 2022;87(2):1133–59. <https://doi.org/10.1007/s10708-020-10302-4>.

33. Elalem S, Pal I. Mapping the vulnerability hotspots over Hindu-Kush Himalaya region to flooding disasters. *Weather Clim Extrem.* 2015;8:46–58. <https://doi.org/10.1016/j.wace.2014.12.001>.
34. Eniolorunda NB, Mashi SA, Nsofor GN. Toward achieving a sustainable management: characterization of land use/land cover in Sokoto Rima floodplain, Nigeria. *Environ Dev Sustain.* 2017;19:1855–78. <https://doi.org/10.1007/s10668-016-9831-6>.
35. Feizizadeh B, Omarzadeh D, Kazemi Garajeh M, Lakes T, Blaschke T. Machine learning data-driven approaches for land use/cover mapping and trend analysis using Google Earth Engine. *J Environ Plan Manag.* 2023;66(3):665–97. <https://doi.org/10.1080/09640568.2021.2001317>.
36. Fortin MJ, Boots B, Csillag F, Remmel TK. On the role of spatial stochastic models in understanding landscape indices in ecology. *Oikos.* 2003;102(1):203–12. <https://doi.org/10.1034/j.1600-0706.2003.12447.x>.
37. Gao C, Cheng D, Iqbal J, Yao S. Spatiotemporal change analysis and prediction of the great yellow river region (GYRR) land cover and the relationship analysis with mountain hazards. *Land.* 2023;12(2):340. <https://doi.org/10.3390/land12020340>.
38. Gorelick N, Hancher M, Dixon M, Ilyushchenko S, Thau D, Moore R. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sens Environ.* 2017;202:18–27. <https://doi.org/10.1016/j.rse.2017.06.031>.
39. Gupta L, Dixit J. A GIS-based flood risk mapping of Assam, India, using the MCDA-AHP approach at the regional and administrative level. *Geocarto Int.* 2022. <https://doi.org/10.1080/10106049.2022.2060329>.
40. Gupta L, Dixit J, Pandey PC, Pandey M. Assessment of forest cover dynamics for the detection of deforestation in the Hindu Kush Himalayan region using geospatial and machine learning approaches. *Earth Sci Inform.* 2025;18(1):160.
41. Hafez A. Evaluation of change detection techniques for monitoring land-cover changes: a case study in the new Burg El-Arab area. *Alex Eng J.* 2011;50:187–95. <https://doi.org/10.1016/j.aej.2011.06.001>.
42. Hasan SS, Alharbi OA, Fahil AS. Land use and land cover mapping using Landsat-8 and Sentinel-2 data in Al Qunfudhah coast, western Saudi Arabia: a comparative study. *Regional Stud Mar Sci.* 2024. <https://doi.org/10.1016/j.rsma.2024.103429>.
43. Hegazy I, Kaloo M. Monitoring urban growth and land use change detection with GIS and remote sensing techniques in Daqahliya governorate Egypt. *Int J Sustain Built Environ.* 2015;4:117–24. <https://doi.org/10.1016/j.ijsbe.2015.02.005>.
44. Henits L, Jürgens C, Mucsi L. Seasonal multitemporal land-cover classification and change detection analysis of Bochum, Germany, using multitemporal Landsat TM data. *Int J Remote Sens.* 2016;37(15):3439–54. <https://doi.org/10.1080/01431161.2015.1125558>.
45. Huang H, Chen Y, Clinton N, Wang J, Wang X, Liu C, et al. Mapping major land cover dynamics in Beijing using all Landsat images in Google Earth Engine. *Remote Sens Environ.* 2017;202:166–76. <https://doi.org/10.1016/j.rse.2017.02.021>.
46. Jia K, Wei X, Gu X, Yao Y, Xie X, Li B. Land cover classification using Landsat 8 Operational Land Imager data in Beijing. *China Geocarto Int.* 2014;29(8):941–51. <https://doi.org/10.1080/10106049.2014.894586>.
47. Jucker T, Caspersen J, Chave J, Antin C, Barbier N, Bongers F, et al. Allometric equations for integrating remote sensing imagery into forest monitoring programmes. *Glob Change Biol.* 2017;23(1):177–90. <https://doi.org/10.1111/gcb.13388>.
48. Kamaraj M, Rangarajan S. Predicting the future land use and land cover changes for Bhavani Basin, Tamil Nadu, India, using QGIS MOLUSCE plugin. *Environ Sci Pollut Res.* 2022;29(57):86337–48. <https://doi.org/10.1007/s11356-021-17904-6>.
49. Kargil, NIC. Agriculture Department | District Kargil, Union Territory of Ladakh | India. URL <https://kargil.nic.in/agriculture-department/> (2025). Accessed 11.20.25.
50. Khan D, Khan N. Decoding urban expansion: a machine learning perspective on Lucknow's growth trajectory. *GeoJournal.* 2025a;90(2):66.
51. Khan D, Khan N. Modelling urban future: integrating CA-ANN model for comprehensive understanding of land use, land cover changes, and temperature dynamics in Lucknow City, India. *Geol Ecol Landsc.* 2025b;1–26. <https://doi.org/10.1080/24749508.2025.2524207>.
52. Khan D, Bano S, Khan N. Spatio-temporal analysis of urbanization effects: unravelling land use and land cover dynamics and their influence on land surface temperature in Aligarh City. *Geol Ecol Landsc.* 2024. <https://doi.org/10.1080/24749508.2024.2409488>.
53. Khan D, Khan N, Ullah S. Harnessing hybrid intelligence and explainable AI for urban growth prediction: A Data-Driven framework for sustainable cities. *Environ Dev Sustain.* 2025;1–40. <https://doi.org/10.1007/s10668-025-06860-7>.
54. Khan D, Khan N, Choudhury U, Singh SK, Kanga S, Kumar P, et al. Urban expansion and spatial growth patterns in Lucknow: implications for sustainable development (1991–2021). *Sustainability.* 2024b;17(1):227.
55. Khawaldah HA. A prediction of future land use/land cover in Amman area using GIS-based Markov Model and remote sensing. *J Geogr Inf Syst.* 2016;8(3):412–27. <https://doi.org/10.4236/jgis.2016.83035>.
56. Kholia T, Kandpal AS. Attitude and constraints faced by the beneficiaries of Pradhan Mantri Krishi Sinchayee Yojana in Garhwal region of Uttarakhand. 2023. <https://www.gbpuat.res.in/paperdetail.php?paper=1452>.
57. Kobayashi N, Tani H, Wang X, Sonobe R. Crop classification using spectral indices derived from Sentinel-2A imagery. *J Inf Telecommun.* 2020;4(1):67–90. <https://doi.org/10.1080/24751839.2019.1694765>.
58. Koroso NH, Zevenbergen JA, Lengoiboni M. Urban land use efficiency in Ethiopia: an assessment of urban land use sustainability in Addis Ababa. *Land Use Policy.* 2020;99:105081. <https://doi.org/10.1016/j.landusepol.2020.105081>.
59. Kumar R. Flood inundation and hazard mapping of 2017 floods in the Rapti River basin using Sentinel-1A synthetic aperture radar images. In: *Applications and Challenges of Geospatial Technology: Potential and Future Trends*; 2019. 77–98. https://doi.org/10.1007/978-3-319-99882-4_6.
60. Kumar R, Acharya P. Flood hazard and risk assessment of 2014 floods in Kashmir Valley: a space-based multisensor approach. *Nat Hazards.* 2016;84(1):437–64. <https://doi.org/10.1007/s11069-016-2428-4>.
61. Kumar N, Ramanathan A, Tranter M, Sharma P, Pandey M, Ranjan P, et al. Switch in chemical weathering caused by the mass balance variability in a Himalayan glacierized basin: a case of Chhota Shigri Glacier. *Hydrol Sci J.* 2019;64:179–89. <https://doi.org/10.1080/02626667.2019.1572152>.
62. Liu X, Huang Y, Xu X, Li X, Li X, Ciais P, et al. High-spatiotemporal-resolution mapping of global urban change from 1985 to 2015. *Nat Sustain.* 2020;3(7):564–70. <https://doi.org/10.1038/s41893-020-0521-x>.
63. Loukika KN, Keesara VR, Sridhar V. Analysis of land use and land cover using machine learning algorithms on Google Earth Engine for Munneru River Basin, India. *Sustainability.* 2021;13(24):13758. <https://doi.org/10.3390/su132413758>.
64. Lu D, Hetrick S, Moran E, Li G. Application of time series Landsat images to examining land-use/landcover dynamic change. *Photogramm Eng Remote Sens.* 2012;78(7):747–55. <https://doi.org/10.14358/pers.78.7.747>.

65. Lukas P, Melesse AM, Kenea TT. Prediction of future land use/land cover changes using a coupled CA-ANN model in the upper Omo–Gibe River Basin, Ethiopia. *Remote Sens.* 2023;15(4):1148. <https://doi.org/10.3390/rs15041148>.
66. MNREGA. Mahatma Gandhi NREGA [WWW Document]. URL https://nrega.dord.gov.in/MGNREGA_new/Nrega_home.aspx (2025). Accessed 11.20.25.
67. Mohajane M, Essahlaoui ALI, Oudija F, Hafyani ME, Hmaidi AE, Ouali AE, et al. Land use/land cover (LULC) using Landsat data series (MSS, TM, ETM+, and OLI) in Azrou Forest, in the Central Middle Atlas of Morocco. *Environments.* 2018;5(12):131. <https://doi.org/10.3390/environments5120131>.
68. Mudi S, Das P. Flood hazard mapping in Assam using sentinel-1 SAR data. In: *geospatial technology for environmental hazards: modeling and management in Asian countries*; 2022. pp. 459–473. https://doi.org/10.1007/978-3-030-75197-5_20.
69. Muhammad R, Zhang W, Abbas Z, Guo F, Gwiazdzinski L. Spatiotemporal change analysis and prediction of future land use and land cover changes using QGIS MOLUSCE plugin and remote sensing big data: a case study of Linyi, China. *Land.* 2022;11(3):419. <https://doi.org/10.3390/land11030419>.
70. Mundia CN, Aniya M. Analysis of land use/cover changes and urban expansion of Nairobi City using remote sensing and GIS. *Int J Remote Sens.* 2005;26(13):2831–49. <https://doi.org/10.1080/01431160500117865>.
71. Naikoo MW, Rihan M, Ishtiaque M. Analyses of land use land cover (LULC) change and built-up expansion in the suburb of a metropolitan city: spatio-temporal analysis of Delhi NCR using Landsat datasets. *J Urban Manag.* 2020;9(3):347–59. <https://doi.org/10.1016/j.jum.2020.05.004>.
72. Nasar-u-Minallah M, Zia S, Rahman AU, Riaz O. Spatio-temporal analysis of urban expansion and future growth patterns of Lahore, Pakistan. *Geogr Environ Sustain.* 2021;14(3):41–53. <https://doi.org/10.24057/2071-9388-2020-215>.
73. Nasiri V, Deljouei A, Moradi F, Sadeghi SMM, Borz SA. Land use and land cover mapping using Sentinel-2, Landsat-8 Satellite Images, and Google Earth Engine: a comparison of two composition methods. *Remote Sens.* 2022;14(9):1977. <https://doi.org/10.3390/rs14091977>.
74. NMSA. National Mission for Sustainable Agriculture [WWW Document]. URL <https://nmsa.dac.gov.in/> (2025). Accessed 11.20.25.
75. NRSC. Annual Land Use and Land Cover Atlas of India. Land Use & Cover Mapping and Monitoring Division, SR&LUM Group, Remote Sensing Applications Area, National Remote Sensing Centre, ISRO, Hyderabad. PPEG No. NRSC-RSA-SR & LUMG-LU&CMD-FEB 2024-TR-0002374-V1.0. 2024. p. 115. https://www.nrsc.gov.in/nrscnew/resources_atlas_LULC.php
76. Olofsson P, Foody GM, Herold M, Stehman SV, Woodcock CE, Wulder MA. Good practices for estimating area and assessing accuracy of land change. *Remote Sens Environ.* 2014;148:42–57. <https://doi.org/10.1016/j.rse.2014.02.015>.
77. Pande CB. Land use/land cover and change detection mapping in Rahuri watershed area (MS), India using the Google Earth Engine and machine learning approach. *Geocarto Int.* 2022;37(26):13860–80. <https://doi.org/10.1080/10106049.2022.2086622>.
78. Pandey M, Pandey PC, Ray Y, Arora A, Jawak SD, Shukla UK, editors. *Advances in remote sensing technology and the three poles*. Hoboken: John Wiley & Sons; 2022.
79. Pandey PC, Pandey M. Highlighting the role of agriculture and geospatial technology in food security and sustainable development goals. *Sustain Dev.* 2023;31(5):3175–95. <https://doi.org/10.1002/sd.2600>.
80. Pandey PC, Koutsias N, Petropoulos GP, Srivastava PK, Ben Dor E. Land use/land cover in view of earth observation: data sources, input dimensions, and classifiers—a review of the state of the art. *Geocarto Int.* 2019;36(9):957–88. <https://doi.org/10.1080/10106049.2019.1629647>.
81. Phiri D, Simwanda M, Salekin S, Nyirenda VR, Murayama Y, Ranagalage M. Sentinel-2 data for land cover/use mapping: a review. *Remote Sens.* 2020;12(14):2291. <https://doi.org/10.3390/rs12142291>.
82. PMFBY. Pradhan Mantri Fasal Bima Yojana—Crop Insurance | PMFBY—Crop Insurance [WWW Document]. URL <https://pmfby.gov.in/> (2025). Accessed 11.20.25.
83. PMKSY. Pradhan Mantri Krishi Sinchayee Yojana [WWW Document]. URL <https://pmksy.gov.in/> (2025). Accessed 11.20.25.
84. Prashanth M, Kumar A, Dhar S, Verma O, Rai SK, Kouser B. Land use/land cover change and its implication on soil erosion in an ecologically sensitive Himachal Himalayan watershed, Northern India. *Front For Glob Change.* 2023;6:1124677.
85. Rahman MTU, Tabassum F, Rasheduzzaman M, Saba H, Sarkar L, Ferdous J, et al. Temporal dynamics of land use/land cover change and its prediction using CA-ANN model for Southwestern coastal Bangladesh. *Environ Monit Assess.* 2017;189:1–18. <https://doi.org/10.1007/s10661-017-6272-0>.
86. Randazzo G, Cascio M, Fontana M, Gregorio F, Lanza S, Muzirafuti A. Mapping of Sicilian pocket beaches land use/land cover with Sentinel-2 imagery: a case study of Messina Province. *Land Basel.* 2021;10(7):678. <https://doi.org/10.3390/land10070678>.
87. Ruas RDB, Costa LMS, Bered F. Urbanization driving changes in plant species and communities—a global view. *Glob Ecol Conserv.* 2022;38:e02243. <https://doi.org/10.1016/j.gecco.2022.e02243>.
88. Raza D, Khushi M, Shu H, Aslam H, Saleem MS, Ahmad A, et al. CA-ANN based LULC prediction and influence assessment on LST-NDVI using multi-temporal satellite images. *Environ Earth Sci.* 2024;83(5):144. <https://doi.org/10.1007/s12665-024-11467-8>.
89. Redowan M, Phinn S, Roelfsema C, Aziz AA. Modeling forest cover dynamics in Bangladesh using multi-layer perceptron neural network with Markov chain. *J Appl Remote Sens.* 2022;16(3):034502–034502. <https://doi.org/10.1117/1.JRS.16.034502>.
90. Regmi RR, Saha SK, Subedi DS. Geospatial analysis of land use land cover change modeling in Phewa Lake watershed of Nepal by using GEOMOD model. In: *Himalayan Physics*; 2017. pp. 65–72.
91. Sadorsky P. The effect of urbanization and industrialization on energy use in emerging economies: implications for sustainable development. *Am J Econ Sociol.* 2014;73(2):392–409. <https://doi.org/10.1111/ajes.12072>.
92. Sardar P, Samadder SR. Understanding the dynamics of landscape of greater Sundarban area using multi-layer perceptron Markov chain and landscape statistics approach. *Ecol Indic.* 2021;121:106914. <https://doi.org/10.1016/j.ecolind.2020.106914>.
93. Sarkar D, Sarkar T, Saha S, Mondal P. Compiling non-parametric tests along with CA-ANN model for precipitation trends and variability analysis: a case study of Eastern India. *Water Cycle.* 2021;2:71–84. <https://doi.org/10.1016/j.watcyc.2021.11.002>.

94. Sekertekin A, Marangoz AM, Akcin H. Pixel-based classification analysis of land use land cover using Sentinel-2 and Landsat-8 data. *Int Arch Photogramm Remote Sens Spat Inf Sci*. 2017;42:91–3. <https://doi.org/10.5194/isprs-archives-XLII-4-W6-91-2017>.
95. Shao Z, Sumari NS, Portnov A, Ujoh F, Musakwa W, Mandela PJ. Urban sprawl and its impact on sustainable urban development: a combination of remote sensing and social media data. *Geo-spat Inf Sci*. 2021;24:241–55. <https://doi.org/10.1080/10095020.2020.1787800>.
96. Shetty, S. Analysis of machine learning classifiers for LULC classification on Google Earth Engine (Master's thesis, University of Twente). 2019.
97. Simelane SP, Hansen C, Munghemezulu C. The use of remote sensing and GIS for land use and land cover mapping in Eswatini: a review. *S Afr J Geomat*. 2021;10(2):181–206. <https://doi.org/10.4314/sajg.v10i2.13>.
98. Simwanda M, Murayama Y. Spatiotemporal patterns of urban land use change in the rapidly growing city of Lusaka, Zambia: implications for sustainable urban development. *Sustain Cities Soc*. 2018;39:262–74. <https://doi.org/10.1016/j.scs.2018.01.039>.
99. Singh SK, Mustak S, Srivastava PK, Szabó S, Islam T. Predicting spatial and decadal LULC changes through cellular automata Markov chain models using earth observation datasets and geo-information. *Environ Process*. 2015;2:61–78. <https://doi.org/10.1007/s40710-015-0062-x>.
100. Singh P, Kala R, Bhavsar D, Roy A, Karnatak H. Urban explosion and hotspots of forest loss in Western Himalaya: mapping land use/cover change trends since 1975. *Adv Space Res*. 2024;74:1238–52. <https://doi.org/10.1016/j.asr.2024.05.015>.
101. Surya B, Ahmad DNA, Sakti HH, Sahban H. Land use change, spatial interaction, and sustainable development in the metropolitan urban areas, South Sulawesi Province, Indonesia. *Land Basel*. 2020;9(3):95. <https://doi.org/10.3390/land9030095>.
102. Taloor AK, Kumar V, Singh VK, Singh AK, Kale RV, Sharma R et al. Land use land cover dynamics using remote sensing and GIS Techniques in Western Doon Valley, Uttarakhand, India. In: *Geoeecology of landscape dynamics*. Singapore: Springer Singapore; 2020. pp. 37–51.
103. Tu Y, Lang W, Yu L, Li Y, Jiang J, Qin Y, et al. Improved mapping results of 10 m resolution land cover classification in Guangdong, China using multisource remote sensing data with Google Earth Engine. *IEEE J Sel Top Appl Earth Obs Remote Sens*. 2020;13:5384–97. <https://doi.org/10.1109/JSTARS.2020.3022210>.
104. Ullah S, Khan M, Qiao X. Evaluating the impact of urbanization patterns on LST and UHI effect in Afghanistan's Cities: a machine learning approach for sustainable urban planning: S. Ullah et al. *Environ Dev Sustain*. 2025;1–42. <https://doi.org/10.1007/s10668-025-06249-6>.
105. United Nations: Department of Economic and Social Affairs. The 2030 Agenda for Sustainable Development. 2015. <https://doi.org/10.1201/b20466-7>.
106. Vaddiraju SC, Talari R, Bhavana K, Apsana S. Predicting the future land use and land cover changes for Saroor Nagar Watershed, Telangana, India, using open-source GIS. *Environ Monit Assess*. 2023;195(12):1499. <https://doi.org/10.1007/s10661-023-12128-2>.
107. Vali A, Comai S, Matteucci M. Deep learning for land use and land cover classification based on hyperspectral and multispectral earth observation data: a review. *Remote Sens*. 2020;12(15):2495. <https://doi.org/10.3390/rs12152495>.
108. Varade D, Sure A, Dikshit O. Potential of Landsat-8 and Sentinel-2A composite for land use land cover analysis. *Geocarto Int*. 2019;34(14):1552–67. <https://doi.org/10.1080/10106049.2018.1497096>.
109. Vashisht P, Pandey M, Ramanathan AL, Tayal S, Jackson M. Comparative assessment of volume change in Kolahoi and Chhota Shigri Glaciers, Western Himalayas, using empirical techniques. *J Clim Change*. 2017;3:37–48. <https://doi.org/10.3233/JCC-170004>.
110. Veldkamp A, Lambin EF. Predicting land-use change. *Agric Ecosyst Environ*. 2001;85(1–3):1–6. [https://doi.org/10.1016/S0167-8809\(01\)00199-2](https://doi.org/10.1016/S0167-8809(01)00199-2).
111. Veloso A, Mermoz S, Bouvet A, Le Toan T, Planells M, Dejoux JF, et al. Understanding the temporal behavior of crops using Sentinel-1 and Sentinel-2-like data for agricultural applications. *Remote Sens Environ*. 2017;199:415–26. <https://doi.org/10.1016/j.rse.2017.07.015>.
112. Vijayaraghavalu SS, Arumugam K, Dange S. Spatio-temporal dynamics of urbanization and environmental sustainability: a predictive modelling approach to forecasting land use transitions in Vellore, India. *Results Eng*. 2025;27:106572. <https://doi.org/10.1016/j.rineng.2025.106572>.
113. Vinayak B, Lee HS, Gedem S. Prediction of land use and land cover changes in Mumbai City, India, using remote sensing data and a multi-layer perceptron neural network-based Markov chain model. *Sustainability*. 2021;13(2):471. <https://doi.org/10.3390/su13020471>.
114. Wang SW, Gebru BM, Lamchin M, Kayastha RB, Lee WK. Land use and land cover change detection and prediction in the Kathmandu district of Nepal using remote sensing and GIS. *Sustainability*. 2020a;12(9):3925. <https://doi.org/10.3390/su12093925>.
115. Wang Y, Li Z, Zeng C, Xia GS, Shen H. An urban water extraction method combining deep learning and Google Earth Engine. *IEEE J Sel Top Appl Earth Obs Remote Sens*. 2020b;13:769–82. <https://doi.org/10.1109/JSTARS.2020.2971783>.
116. Wickham J, Stehman SV, Sorenson DG, Gass L, Dewitz JA. Thematic accuracy assessment of the NLCD 2016 land cover for the Conterminous United States. *Remote Sens Environ*. 2021;257:112357. <https://doi.org/10.1016/j.rse.2021.112357>.
117. Wu KY, Ye XY, Qi ZF, Zhang H. Impacts of land use/land cover change and socioeconomic development on regional ecosystem services: the case of fast-growing Hangzhou metropolitan area, China. *Cities*. 2013;31:276–84. <https://doi.org/10.1016/j.cities.2012.08.003>.
118. Yadav S, Lama S, Kapley A. Urban Environmental Issues and Mitigation by Applying Ecological and Ecosystem Engineering. In: *Handbook of ecological and ecosystem engineering*. 2021. pp. 123–145. <https://doi.org/10.1002/9781119678595.ch6>.
119. Yang J, Li S, Lu H. Quantitative influence of land-use changes and urban expansion intensity on landscape pattern in Qingdao, China: Implications for urban sustainability. *Sustainability*. 2019;11(21):6174. <https://doi.org/10.3390/su11216174>.
120. Zhang C, Di L, Yang Z, Lin L, Hao P. AgKit4EE: a toolkit for agricultural land use modeling of the Conterminous United States based on Google Earth Engine. *Environ Model Softw*. 2020;129:104694. <https://doi.org/10.1016/j.envsoft.2020.104694>.
121. Zhang L, Xu M, Chen H, Li Y, Chen S. Globalization, green economy and environmental challenges: state of the art review for practical implications. *Front Environ Sci*. 2022;10:870271. <https://doi.org/10.3389/fenvs.2022.870271>.

122. Zhao G, Lin G, Warner T. Thematic mapper data for change detection and sustainable use of cultivated land: a case study in the Yellow River Delta, China. *Int J Remote Sens.* 2004;25(13):2509–22. <https://doi.org/10.1080/01431160310001619571>.
123. Zhao Z, Islam F, Waseem LA, Tariq A, Nawaz M, Islam IU, et al. Comparison of three machine learning algorithms using Google Earth Engine for land use land cover classification. *Rangeland Ecol Manage.* 2024;92:129–37. <https://doi.org/10.1016/j.rama.2023.10.007>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.